COURSE
OVERVIEW

FALL ARREST & BASIC RESCUE

This course provides learners with the knowledge and practical understanding required to implement an approved Fall Protection Plan on site. It focuses on legal compliance, risk assessment, medical fitness, training requirements, equipment management, rescue planning, and ongoing review processes. The course equips appointed supervisors and implementers to ensure that all work at height activities are conducted safely, systematically, and in accordance with legislation, industry best practice, and company requirements.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

Implement a Fall Protection Plan on site

Identify and control fall risks, verify medical fitness and training compliance,

Manage fall protection equipment, ensure rescue preparedness,

Communicate safety requirements effectively, and monitor,

Review, and report on fall protection controls to maintain a safe work at height environment.

INTRODUCTION

GRAVITY VALUES

R — RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.

I — INTEGRITY

Conducting day-to-day business with integrity at all times.

S — SOLUTIONS **DRIVEN**

Being solution driven.

Q — QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

Before any work begins, the site must be safe.
Use this simple checklist

Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

01

Rescue Plan

Have a rescue plan ready.
A rescue kit and first aid kit must be on site.

02

Work in Pairs

There must be at least two workers on site.
If something goes wrong, one can help or call for help.

03

Check Equipment

All tools and gear must be safe and working before use.

04

Safe Site

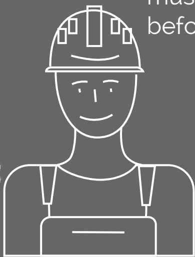
05

Check the building or structure.

Make sure it is strong and safe to work on.

SITE ESTABLISHMENT

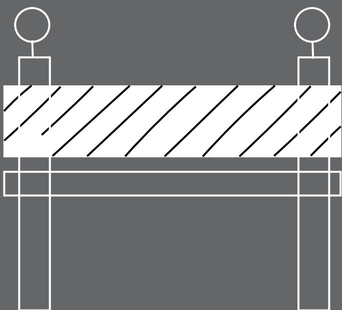
SITE ESTABLISHMENT



06

Manage Traffic

If there are cars or trucks nearby, there must be a traffic safety plan.



07

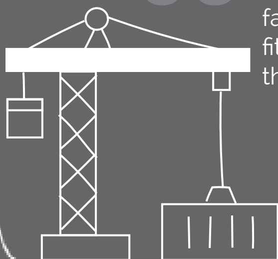
Do a Risk Assessment

Check what dangers there are. All workers must understand the risks before starting work.

08

Use the Right Safety Gear

Make sure the fall protection fits the job and the site.



EXCLUSION ZONES **& BARRICADES**

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY

Be healthy and ready before starting work.

Only do jobs you are trained and allowed to do.

Never work under the influence of alcohol or drugs.

Always wear your safety gear (PPE, harness if working at height).

Use harnesses correctly and stay connected at heights.

Never work alone and always follow safety steps.

SITE SAFETY

- **Keep people away** from areas below when working above.
- **Don't walk under lifted loads** or stand in dangerous areas.
- **Only trained people** can do electrical, underground, or confined space work.
- **Follow road rules.** Wear helmets. No passengers on open trucks..
- **Don't drive tired** or distracted.
- **Stop unsafe work!** Everyone has the right to speak up.

GRAVITY

A. Fall Protection Plan

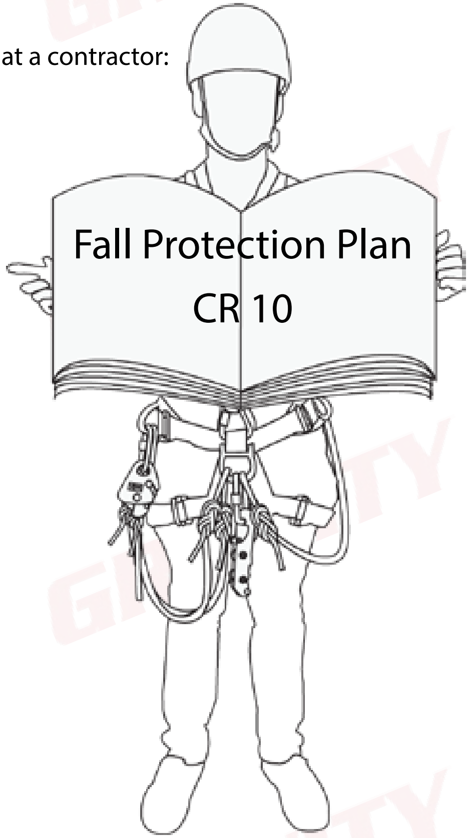
1. Introduction to Fall Protection Plan

Before any work at height may commence, all the fall risks on site must be addressed and documented in a fall protection plan (FPP).

Construction Regulations (CR) of 2014, Section 10 (CR10) sets out all the requirements for such a plan.

The purpose of CR10 of 2014 is to ensure that a contractor:
- Identifies,
- Evaluates, and
- Mitigates

all prevalent fall risks and hazards on site.



According to CR10(2) the Fall Protection Plan must include:

- CR10(2)(a) - **Risk Assessment** - Forms the foundation of the plan.
- CR10(2)(b) - **Medical Fitness -** All workers must be medically fit, and reviewed once a year.
- CR10(2)(c) - A **Training Programme** - All workers must receive relevant training.
- CR10(2)(d) - **Equipment Management** - All equipment must be "safe for use and fit for purpose". The equipment manager must be trained in the basic requirements of equipment management.
- CR10(2)(e) - **Rescue Plan** - A trained rescuer must be able to rescue a worker from any situation.

An FPP must also be reviewed after any changes on site, or should the work scope change.

Note: All this must be done BEFORE work may start.

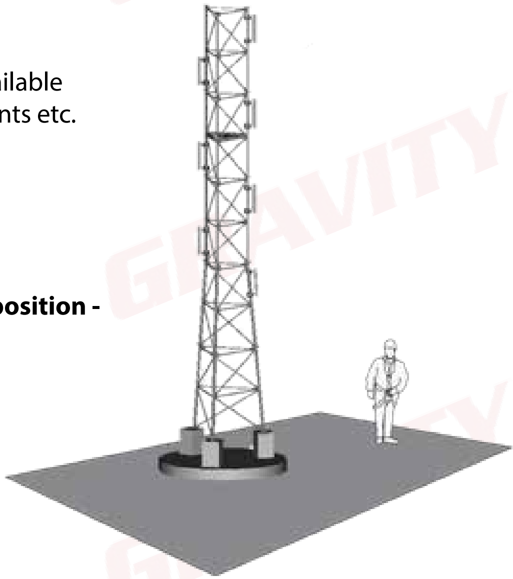
2. Risk Assessment

The most important part of the Fall Protection Plan is the **Risk Assessment.**

STEP 1 : IDENTIFY POSSIBLE RISKS

What to do when approaching a worksite

BEFORE any work may start on a site the following must be identified:



- **Site requirements -**
  - What type of structure is available
  - What are the PPE requirements etc.
  - Permit to work
- **Access method requirements -**
  - Fall Arrest
  - Rope Access
  - Other
- **What is the required team composition -**
  - How many team members
  - Who must do what
  - Who will be who's buddy
- **Fall protection requirements -**
  - Work Restraint
  - Fall Arrest
  - Lifelines etc.
- **PPE requirements -**
  - Shock absorbing lanyards
  - Backup devices
  - Descending devices etc.
- **Scope of work**
- **Tools required for work**
- **Possible risks on site that might cause a fall**
  - e.g. bad weather, dangerous animals etc.

After this, the risks must be communicated to everyone on site by means of:

- Site induction
- Toolbox talk

With this information we can now prepare a Risk Assessment.

Below is an example of a risk assessment. Note that this is only an example and it is of utmost importance that the PRINCIPLES behind successful risk mitigation are understood and not just the format of the document.

When a risk is identified, it needs to be analysed and given a Risk Rating - it is given a rating twice - once before any control methods have been implemented and once afterwards. In order for us to consistently analyse our identified risk and ensure that we can reproduce any rating, we need a "documented method" for this process as stated in CR 9(1)(c).

STEP 2 : PRE-MITIGATION RISK RATING

		HAZARD SEVERITY				
		Negligible	**Slight**	**Moderate**	**High**	**Very High**
LIKELIHOOD OF OCCURRENCE	**Very Unlikely**	*LOW*	*LOW*	*LOW*	*LOW*	*LOW*
	Unlikely	*LOW*	*LOW*	*LOW*	*MEDIUM*	*MEDIUM*
	Possible	*LOW*	*LOW*	*MEDIUM*	*MEDIUM*	*HIGH*
	Likely	*LOW*	*MEDIUM*	*MEDIUM*	*HIGH*	*HIGH*
	Very Likely	*MEDIUM*	*MEDIUM*	*HIGH*	*HIGH*	*HIGH*

No work is allowed to commence if there is a "**Medium**" or "**High**" risk present on site.
Thus, we have to mitigate all identified risks in order for them to receive a "**Low**" risk rating.

STEP 3 : RISK MITIGATION HIERARCHY

This is a process to help a company develop the correct procedures or methods to manage the risks in a reasonable and practical way.

STEP 1: Avoid the risk completely
Do not do the work; this will prevent the worker from being exposed to any risk.

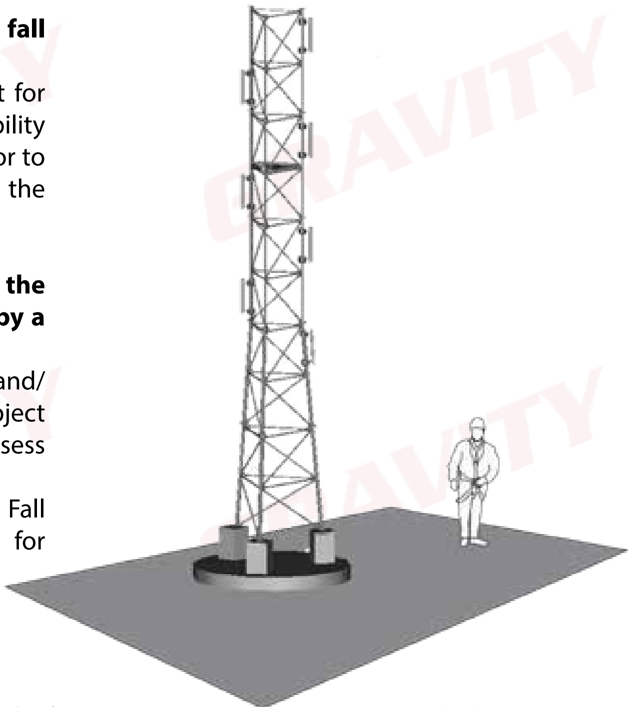
STEP 2: Isolate the risk
Prevent workers from entering the area where there is a fall risk, **for example:** erect a solid barrier or handrail so that there is no risk of falling.

STEP 3: Engineer the risk away
Eliminate the need to access the fall risk area, **for example:** by locating or relocating items elsewhere if they require inspection and maintenance. Provide alternate means of access (which removes the fall risk), **for example:** walkways and telescopic handles or stairways instead of ladders etc.).

STEP 4: Use fall prevention systems
Use suitable fall restraint systems and PPE that will prevent workers from reaching an edge where a fall may occur.

STEP 5: Select appropriate fall arrest system
Which is "safe for use and fit for purpose". Ensure the availability of a competent site supervisor to identify access method and the correct rescue plan.

STEP 6: Stop work until the risks have been mitigated by a competent person
Consult the Site Supervisor and/or a work at height Subject Matter Expert (SME) and reassess the worksite and rescue plan.
Alternatively, contact the Fall Protection Plan Developer for advice.



Step 6 can cause some controversy, so here are a few guidelines on when to stop work:

- **Adverse Weather Conditions:**
  No work-at-height shall be done during any inclement weather conditions unless all risks have been mitigated adequately. Should the working situation change whilst working at height, all technicians must safely descend to safety and the Site Supervisor must report back to the Project Manager immediately. Examples of inclement weather conditions:
  - Rain
  - Lightning
  - Extremely cold weather
  - Extremely hot weather
  - Sleet or snow
  - High or gusty wind
  - Poor visibility (fog or smog)

  When working in windy conditions, the Beaufort Wind Force Scale can be used to determine the safety of the worksite. The Site Supervisor must ultimately take responsibility and work can be done if the risk assessment allows the working team to continue. Wind conditions must be continually monitored during the task as the conditions may change.

- **Environmental Factors:**
  Apart from weather, any environmental factor that poses a danger to a worker must be addressed before work may begin. If this factor only becomes present after work has already begun, work has to be stopped until the risk has been mitigated. Examples of this include bees, snakes, thorn bushes, gasses, poor visibility, etc.

  No work-at-height shall be done at night, unless adequate lighting is available on site, such as head-lamps or floodlights. Any hazards pertaining to lighting must be mitigated accordingly in the risk assessment.

- **Breach of Risk Assessment:**
  If, at any time, any person on site operates outside of the risk mitigations agreed upon during the toolbox talk, work must halt immediately and the person must either be removed from site (if for example it is a member of the public), or the toolbox talk must be repeated and explained again. This includes, but is not limited to:
  - A person working without correct PPE.
  - A person who is not medically fit.
  - A person who has not been adequately trained etc.

STEP 4 : POST MITIGATION RISK RATING

After the mitigation, the risk must receive another "risk rating".
Now they MUST be "**Low**"; if not, the process must be repeated.

		HAZARD SEVERITY				
		Negligible	Slight	Moderate	High	Very High
LIKELIHOOD OF OCCURRENCE	Very Unlikely	*LOW*	*LOW*	*LOW*	*LOW*	*LOW*
	Unlikely	*LOW*	*LOW*	*LOW*	*MEDIUM*	*MEDIUM*
	Possible	*LOW*	*LOW*	*MEDIUM*	*MEDIUM*	*HIGH*
	Likely	*LOW*	*MEDIUM*	*MEDIUM*	*HIGH*	*HIGH*
	Very Likely	*MEDIUM*	*MEDIUM*	*HIGH*	*HIGH*	*HIGH*

Class Exercise 1

Complete the Risk Assessment below

Hazard	Pre-Mitigation Risk	Tick	Mitigation	Tick	Post-Mitigation Risk	Tick
Falling from Height	High		Correct PPE		High	
			Always Work Alone			
	Medium		Correct Rescue Procedures		Medium	
			Work in Teams of at Least Two (Buddy System)			
	Low		Diving Instruction		Low	
			Work at Height Training			

B. Work at Height Basics

1. Shock Load

When an object is falling, it gathers speed, and as it gathers speed, it gathers momentum. When this object stops falling, the energy it possesses due to its momentum needs to be directed somewhere; usually this would be the ground.

Newton's Third Law assures us that for every action there is an equal, but opposite reaction. The energy transferred from the object to the ground is in turn given back. If all of this sounds confusing, think of it like this:

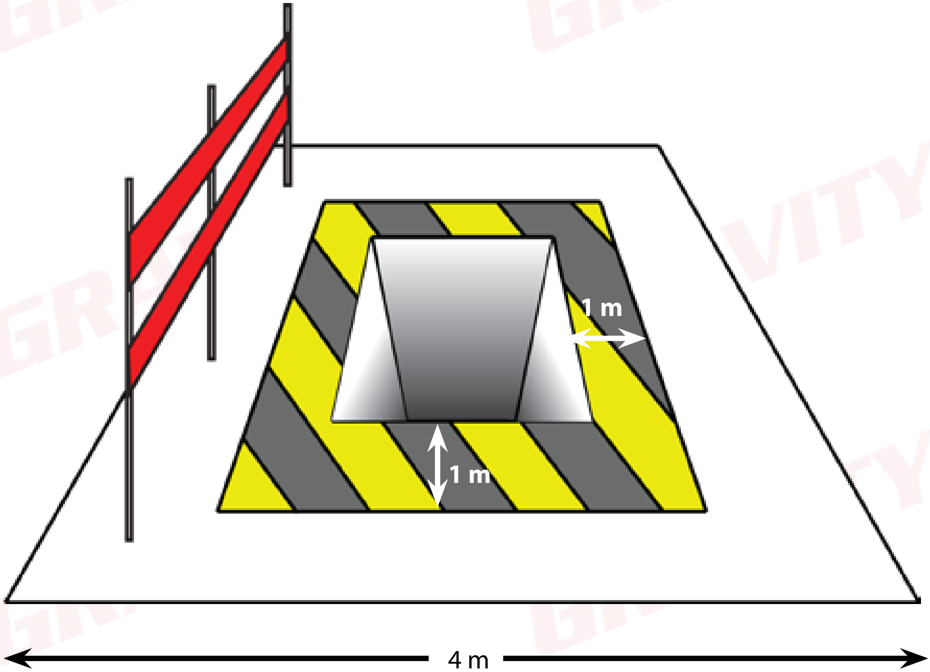
A man standing on a chair, jumps off and DOES NOT bend his knees. After jumping off the chair and not bending the knees, there is no absorption of the shock load, in the same way as when using a non-energy absorbing lanyard to arrest a fall. This can cause serious injury or even death if the fall is from a greater height.

When bending the knees upon hitting the ground, the person's legs absorb the energy generated during the fall by increasing both the time it takes to come to a stop, as well as the distance. This is the same way an energy-absorbing lanyard would work.

2. Exclusion Zones and Barricading

The easiest way to prevent a fall/shock load is by preventing workers from reaching a fall risk position. This can be done by one of two methods:

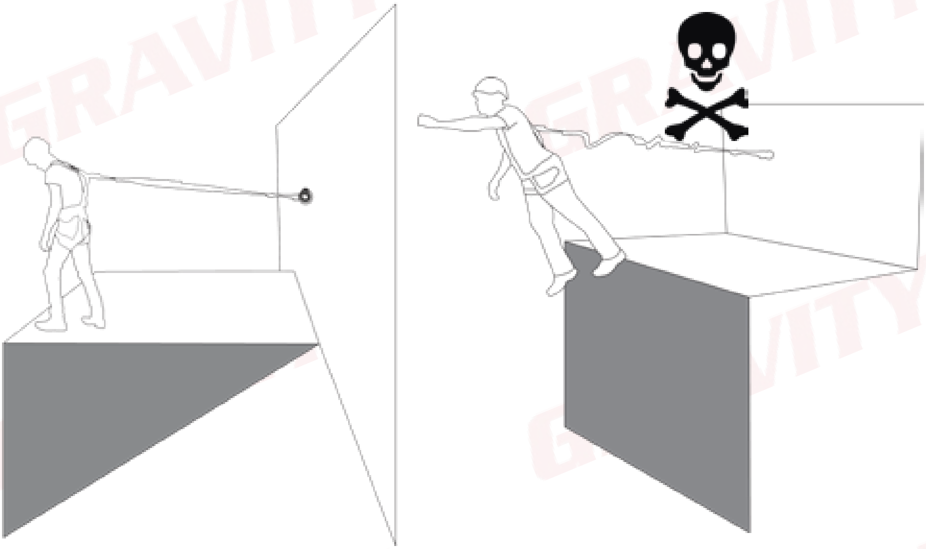
- **Exclusion zones:** A zone or space where people are **not allowed** to walk or work within, for examples, a 1m space around a hole in a roof.
- **Barricading**: A physical barrier **preventing** people from reaching a fall risk position.



Should both of the above mentioned methods not be possible for whatever reason, a person may be made safe by use of a work restraint system, or fall arrest.

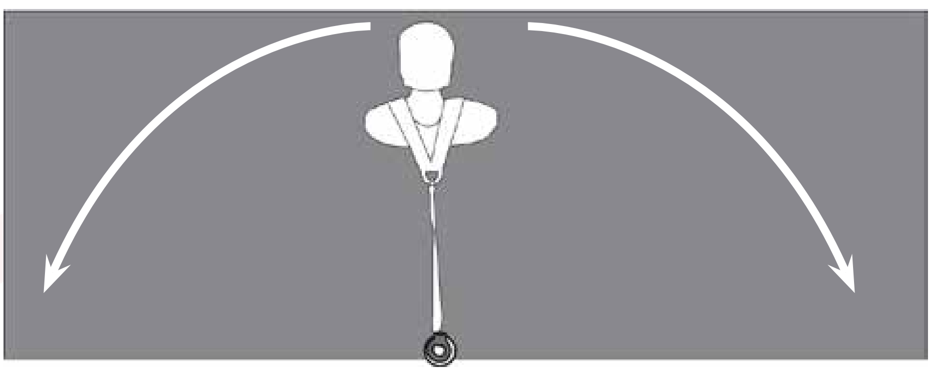
3. Work Restraint

Definition: A system where we physically tie a worker to the structure, thus preventing him/her from reaching the edge, thus preventing a fall.



Advantages: Inexpensive, effective, little training required and minimal equipment required, no shock load, simple rescues .
Disadvantages: Technician's movement is limited.

Remember to ensure that there is no way that you can reach any of the edges surrounding the work area.

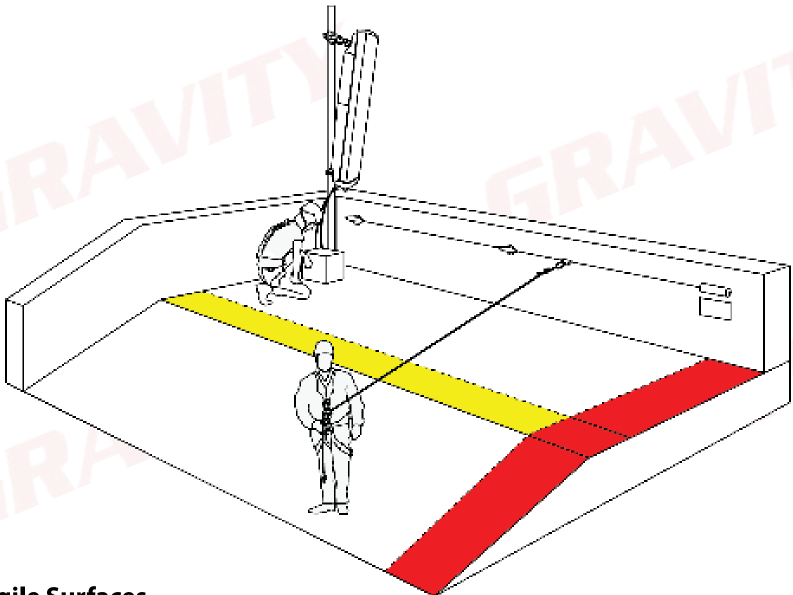


4. Rooftop Work Restraint System

One can combine the horizontal lifeline with a work restraint system to create a system ideal for rooftop work.

This allows a technician to move left to right and the technician can manually manage the distance to reach the required working area.

Special care should be taken when building the rooftop work restraint system to prevent the person from reaching the edge of the work area.



Fragile Surfaces

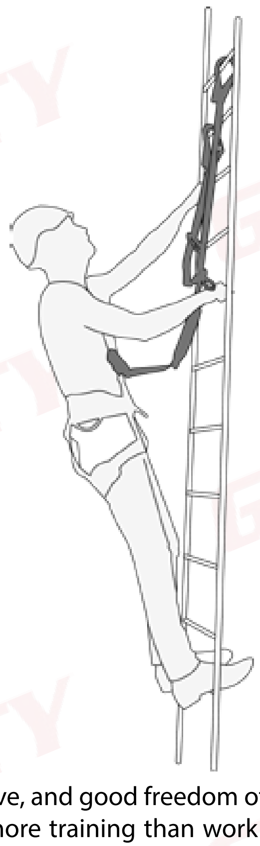
An important part of rooftop work is to remember is that there is always a risk that the roof has been neglected or damaged over time, leading to the surface becoming fragile, which can cause a worker to fall through the roof. To prevent this, ensure that the roof is inspected by the relevant responsible person and that you are permitted to work upon it. This must be covered in your risk assessment before work starts.

5. Fall Arrest

Definition: a system used to suspend a person safely AFTER a fall has occurred, and doing all this in such a way as to prevent injury as far as possible.

Fall Arrest does NOT prevent a fall like Work Restraint. It only prevents a person from serious injury or hitting the ground due to a fall.

A basic Fall Arrest system consists of a physical connection between the worker and the structure, but with some sort of shock absorbing component. Otherwise, it will be a piece of equipment designed in such a way that the worker's fall is arrested so quickly that there is no time to generate a notable shock load.



Advantages: Cheap, effective, and good freedom of movement.
Disadvantages: Requires more training than work restraint, chance of shock load, rescues not as simple as with work restraint.

Now, for us to work safely, and to be able to use either a work restraint or fall arrest system, we need our Personal Protective Equipment or PPE.

6. Rope Access

Definition: A form of work positioning, using a series of connection points and ropes to enable access to difficult to reach locations where other safer methods are not possible.

Much like Fall Arrest, Rope Access does NOT prevent a fall like Work Restraint does. It only prevents a person from seriously getting hurt or hitting the ground due to a fall. Rope Access also employs a double rope system that reduces the likelihood of a ground fall greatly.

2-Point Principle

A rope access technician must at all times have two points of connection, i.e. one point on each of the two ropes. Failure to do so will put the technician in grave danger of a ground fall.

Advantages: Access to difficult positions, less expensive than alternative methods (scaffold, MEWPS, etc.).
Disadvantages: Requires more training.

Now, for us to work safely, and to be able to use either a work restraint or rope access system, we need our Personal Protective Equipment or PPE.

7. Full Range of Personnel Protective Equipment (PPE)

In order for a user to use PPE safely and correctly, there are four things each user must know about each piece of equipment, namely:

- **Identification** - The name and purpose of each piece of equipment.
- **Certification** - Only equipment with the correct South African National Standards (SANS) or European National Standards (EN) should be used and must be purchased through a reputable retailer such as Gravity Training. For example, Gravity Training only makes use of ISO 9001:2015 certified equipment suppliers.
- **Limitation** - How to use and not use equipment, the safe use criteria and hazards to identify when using each piece of equipment, and the maximum load that each piece of equipment is certified to carry. Equipment load bearing ability is indicated as **Breaking Strength** and **Safe Working Load** which should never be exceeded.
- **Inspection** - Pre-use inspection should be done before start of work and that equipment should be sent to the appointed equipment controller for detailed 3 monthly inspections. (This will only be covered in a generic fashion on this course.)

As stated before, a technician must always comply to the 2-point principle, thus the PPE list that follows states how to achieve this with the use of PPE.

Climbing Helmet (EN 12492)

A helmet is used to protect the technician's head during a fall as well as from falling objects:

- Must be rated for top AND side impacts.
- Must be fitted with a rated chin-strap:
  - The chin strap is fitted to keep the helmet on the technician's head during a fall, side impacts and high wind conditions.
- Helmet must be used within the approved temperature ratings of -30 to 50°c.
- The chin-strap breaks at 50 kg to prevent chocking.

Full Body Harness (EN 361, EN 358, EN 813)

A harness allows all other devices and systems to be connected to the worker. This forms the basis of all safety systems.

- All fall arrest systems must be connected to the back D-link or preferably the chest D-link.
- The two buckles on the side of the harness are for work positioning systems.
- The harness must be fitted correctly and with all straps tightened properly.
- Ensure that the buckles go to the correct corresponding connectors.

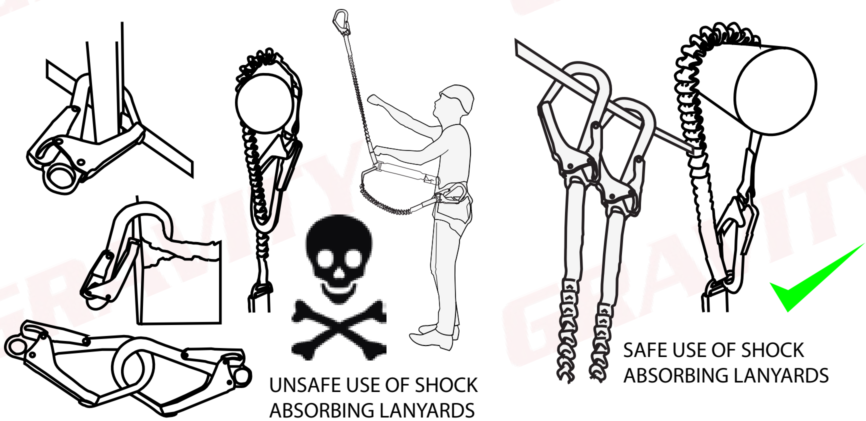
- **Safe Working Load : 150 kg**
- **Breaking Strength : 1500 kg**

Shock Absorbing Lanyard (EN 355)

A shock absorber will catch a worker, should he/she fall, and rip open; this will absorb the shock load (just like with bending the knees when jumping of a chair), thus preventing injury due to shock loading a person's body.

- Primary safety when working at height.
- Shock absorber rips open (deploys) to reduce the shock load felt by the person falling.
- Must be intact during use. Once activated, the shock absorber may not be used again as it will not be able to absorb the shock load of another fall.
- Follow the safe use of connectors below.

- **Safe Working Load : 100 kg**



UNSAFE USE OF SHOCK ABSORBING LANYARDS

SAFE USE OF SHOCK ABSORBING LANYARDS

Work Positioning Lanyard (EN 358)

A work positioning lanyard is used to allow a worker to use both hands freely in order to do work effectively.

- MUST NEVER BE USED WITHOUT A FALL ARREST SYSTEM.

- Should be wrapped around the structure to prevent slipping.

- **Safe Working Load : 150 kg**
- **Breaking Strength : 1500 kg**

The helmet, harness, shock absorber and work positioning lanyard form the minimum PPE that must be used by a worker when at height.

Ascenders (EN 567)

Ascenders refer to any device that allows a worker to climb UP on a rope. They all have a cam with downward facing teeth that bite into the rope, thus preventing them from moving downward. They are divided into two groups, namely:

Handled Ascender (Jumar)

- Must always be closed - on and off the rope.
- Never shock load an ascender as this can cause rope damage.

- **Safe Working Load : 100 kg**
- **Breaking Strength : 1000 kg**
- **Rope Damage : 400 kg**

Chest Ascender (Croll)

- Must always be closed - on and off the rope.
- Never shock load an ascender as this can cause rope damage.
- Ensure that the chest ascender is tightly secured to the harness.

- **Safe Working Load : 100 kg**
- **Breaking Strength : 1000 kg**
- **Rope Damage : 400 kg**

THE CHEST ASCENDER AND HANDLED ASCENDER MUST ALWAYS BE USED TOGETHER IN ORDER TO CREATE 1 POINT FOR THE WORKER.

Descending Devices (EN 341)

Descenders refer to any device that allows a worker to move DOWN on a rope.

Camp Giant

- A descender creates 1 full point on a rope.
- Connects to the bottom D-Link.
- Karabiner must be fitted correctly.
- Must always be locked when not in use.
- Remember to remove all slack before locking off a descender.

- **Safe Working Load : 150 kg**
- **Breaking Strength : 1500 kg**

Guided Type Fall Arrester (including a flexible anchor line) (EN 353-2)

Rope grab refers to any device that can move freely up and down a rope, but will lock onto the rope should a person experience a fall or rapid descent. These devices are used as backup devices to prevent the worker from falling to the ground should the main line fail.

Camp Goblin

- Must be used in a safe Fall Factor.
- Used on a vertical lifeline.
- Used as backup device during rope access.
- Connects to a locker sling.
- Use with a mouse-tail.

- **Safe Working Load : 150 kg**
- **Breaking Strength : 1500 kg**

Cowtails (EN 892)

Cowtails refer to three separate lengths of dynamic rope which are connected to the harness and allows the worker to connect a range of devices onto him/herself.

- Dynamic rope stretches between 20% and 25%.
- Connected with a Figure 8 Follow Through to the harness.
- Make a Figure 8 on a Bight on the other end.
- Ensure all knots are dressed correctly and neatly.
- Ensure that the cowtails are the correct length for the task.

- **Safe Working Load : 250 kg**
- **Breaking Strength : 2500 kg**

Retractable Lanyard (EN360)

Used as a self-retracting fall arrest device, this allows quick and easy movement up and down. The device limits falling distance to approximately 30 cm.

- Similar functionality of a car safety belt, it allows a technician to freely move up and down, and locks only when a person falls.
- Can be used for work positioning and fall arrest.
- Some models include a shock-absorbing function.
- Has a 20 degree maximum range during vertical use.
- Never use above a Fall Factor 1.

Work Rope (EN1891)

Work rope refers to the SEMI-STATIC rope that is used for main and backup lines. Semi-static rope stretches only up to 10%. This limits falling distance, but does not do much for absorbing shock load.

- Stretches between 5% and 10%
- Is of Kernmantle construction and consists of two parts:
  Kern (Core) = 80% of the strength,
  Mantle (sheath) = 20% of the strength.

- **Safe Working Load : 250 kg**
- **Breaking Strength : 2500 kg**

Ribbon stating date of manufacture

Kern\Core

Sheath\Mantle

Open Round Slings (En 566 & EN 795 - Type c)

Used to anchor systems to structures when sharp edges are not present.

- Should be wrapped around the structure to prevent slipping.
- Stitching must be straight to prevent the weakening of the sling.
- Should NOT BE CHOKED.

- **Safe Working Load : 220 kg**
- **Breaking Strength: 2200 kg**

Anchor Slings (EN 566 & EN 795)

Used to anchor systems to structures when sharp edges are present.

- Has PVC covering to protect against sharp edges.
- Must be wrapped around the structure to prevent slipping.
- Connectors must be attached to both buckles.

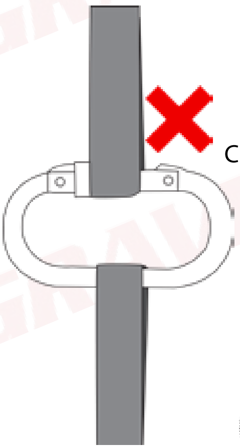
- **Safe Working Load : 220 kg**
- **Breaking Strength : 2200 kg**

Connectors
Karabiners (EN 362)

Used to connect a range of devices and different pieces of equipment.

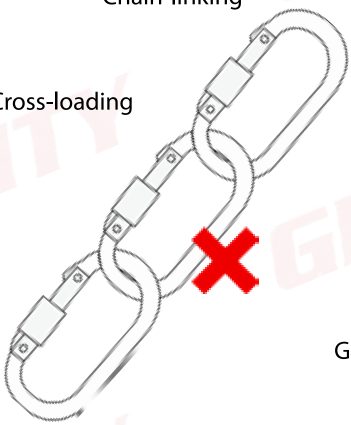
- Always lock karabiners.

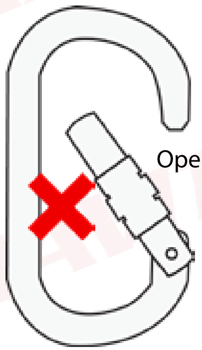
- **Safe Working Load : 220 kg**
- **Breaking Strength : 2200 kg**



Cross-loading

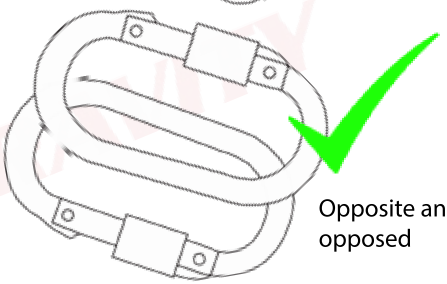
Chain-linking



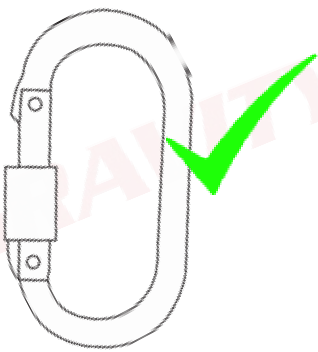


Open gate

Gate facing downward



Opposite and opposed



Etrier

An etrier, or "step-loop", is used alongside the handled ascender in order for the worker to be able to move the chest ascender upwards in order to move up a rope.

Etriers are not load bearing, thus they do not have ratings or weight specifications.

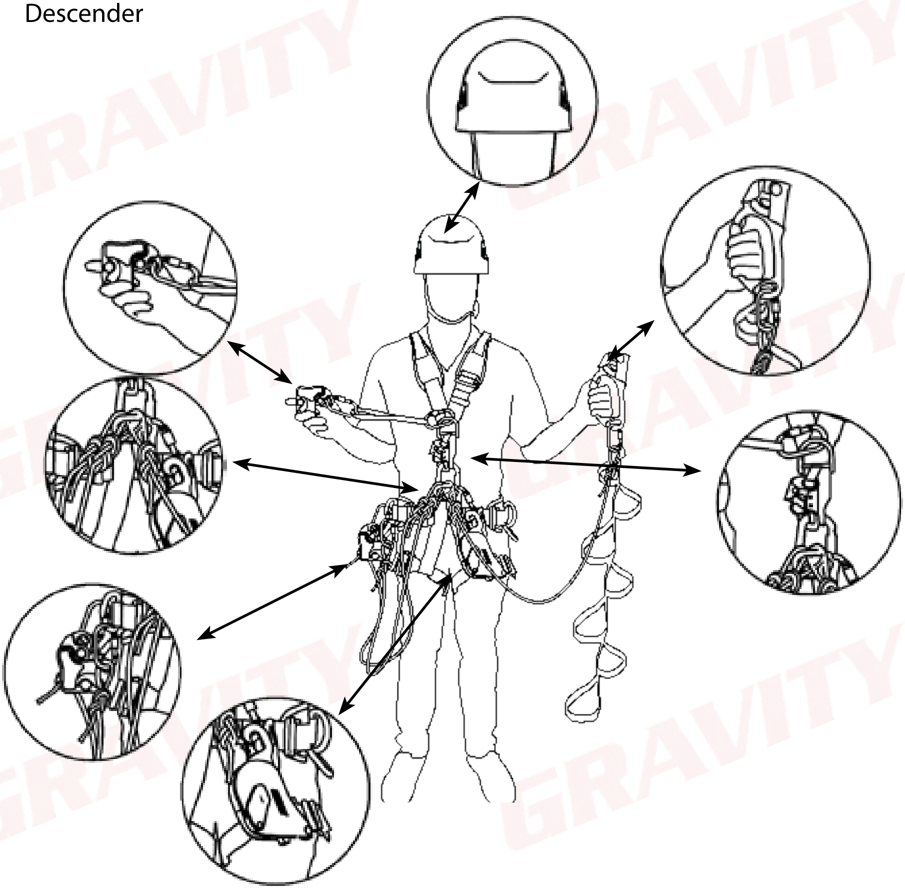
However, it is important that they remain in a good condition; if they were to break while at height, one will have a difficult time removing one's chest ascender.

Correct Assembly

Shown below is the correct assembly of a rope access harness; there is no particular order in which to assemble a harness. A worker must ensure that everything is safe for use and fit for purpose. Always ensure that a "buddy check" before starting work to ensure that everything is safe for use and fit for purpose.

The basic Rope Access PPE is as follows:

- Helmet
- Harness
- 3 Cowtails
- 5 Stainless Steel Karabiners
- 1 Aluminium Karabiner
- Descender
- Backup Device
- Locker Sling
- Handled Ascender
- Chest Ascender
- Etrier (Step-loop)



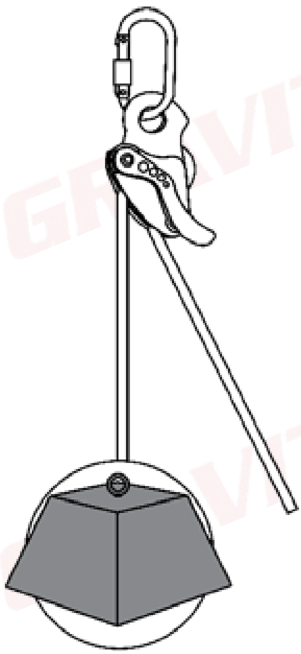
Carrying Tools at Height

When using tools at height it is of utmost importance that all tools are tied off with tool lanyards or accessory cords. This will prevent tools being dropped from height which can cause damage, injury, and even death to people and property below.

When tools that need to be lifted to height weigh more than what the manufacturer recommends, but less than 20 kg, they may not be carried on the harness. It must rather be hoisted up on a separate rope and pulley system. Any equipment weighing more than 20 kg needs to be lifted by a qualified rope rigger.

Principles of Rigging
- Never stand underneath a load.
- Always ensure that the path of the load is clear.
- Ensure that the correct equipment is used for the lift.
-

Principles of Slinging
- Never exceed the safe working load of equipment.
- Always protect slings from sharp edges.
- Always ensure that the load is as secure in the air as it would be on the ground.
- Always consider the center of gravity.



The Weakest Link

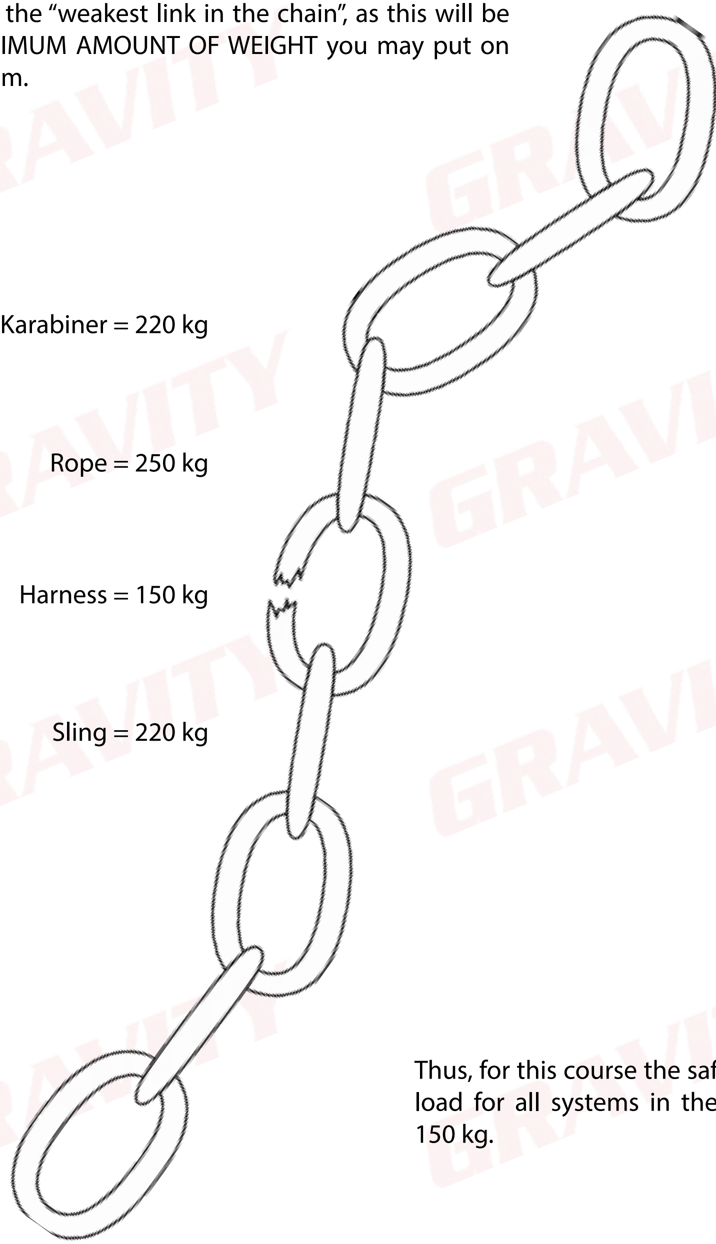
When combining multiple pieces of equipment, always consider the "weakest link in the chain", as this will be the MAXIMUM AMOUNT OF WEIGHT you may put on the system.

Karabiner = 220 kg

Rope = 250 kg

Harness = 150 kg

Sling = 220 kg



Thus, for this course the safe working load for all systems in the course is 150 kg.

Pre-use Inspections

Users must inspect all the equipment to be used before work starts. This must be done to ensure that all equipment is safe for use and fit for purpose in accordance with ISO 22846-2.

Equipment is divided into 2 categories namely hardware and software, each of these have their own inspection criteria. Some pieces of equipment (e.g a harness) fall into both categories and must be inspected as such.

Software

Hardware

Check for:

- Date of manufacture
- Condition of stitching
- Discolouration
- Cuts
- Burns
- Wear and tear
- Certification

Check for:

- Rust
- Working action
- Movement
- Wear and tear
- Deformation
- Cracks

Equipment Storage

Equipment must be stored correctly and according to manufacturer's specifications to prevent damage to equipment when not in use.

Store Equipment:

- In a cool, dry place - This will help prevent mildew in software and rust in hardware.
- Free of chemicals - This will reduce the chance of chemical damage to equipment.
- Where there are no infestations (rats and moths etc.), which can damage software; ensure that there is a system in place to prevent this.
- In a secure location - This will prevent unauthorized personnel entering and potentially damaging equipment.

Rope Care

- Do not use rope for anything other than work at height.
- Do not run rope over sharp edges; use edge protection.
- Do not stand on or walk over a rope.
- Ensure that all devices are compatible with any devices to be used.

Rope Bagging:

It may be preferable to bag a rope instead of coiling it. In such cases, it is recommended to deviate the rope over the structure or ladder and pull the rope into the bag, this will prevent knotting and allows a rope to be bagged faster.

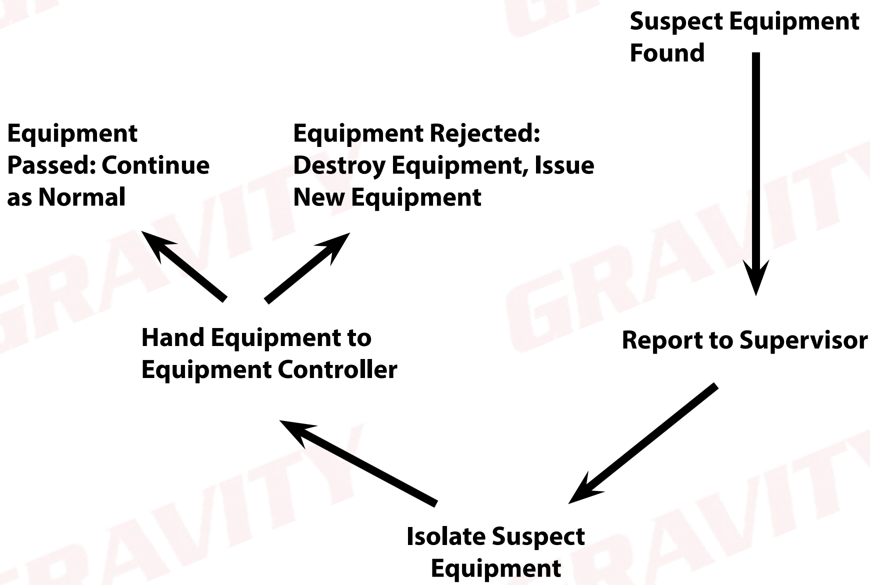
Housekeeping:

Whilst on site, it is important to ensure that the equipment being used is handled in such a way that they do not suffer any unnecessary damage or exposure to environmental conditions. When equipment is not in use , store them in their bags or bins. Ensure that equipment is moved out of the areas where workers move to prevent slips and trips or other accidents.

Reporting Suspect Equipment

Gear that is found to be damaged or too old for use must be reported to the person responsible for the site (either the Site Supervisor or Safety Officer etc).

Damaged or expired equipment must be properly marked and removed from service immediately. Damaged equipment must be destroyed at the earliest opportunity to prevent the equipment from getting back into the work system.



Repairing Damaged Equipment

Equipment may only be repaired by either the manufacturer or a company certified by the manufacturer. **NO CONTRACTOR MAY REPAIR PPE.**

D. Height Safety Principles

1. Fall Factors

"Fall Factors" is the system we use to determine the severity of a fall, or how bad a fall was or will be.

There exists a formula to determine a Fall Factor:

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

There are three main Fall Factors:

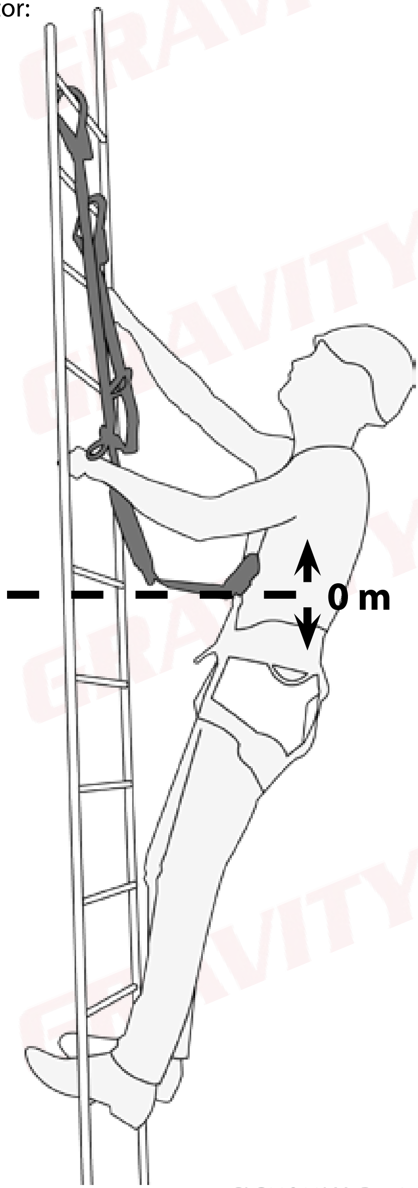
Fall Factor 0
Fall Factor 1
Fall Factor 2

Fall Factor 0: ✔✔

This is the ideal position to be in.

Hooking up as high as possible above the head will result in:

- Almost no falling distance.
- Negligible shock load.
- No deployment of the shock absorber.
- No injury.



$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{0\ \text{m}}{2\ \text{m}}$$

$$\text{Fall Factor} = 0$$

Rope Access

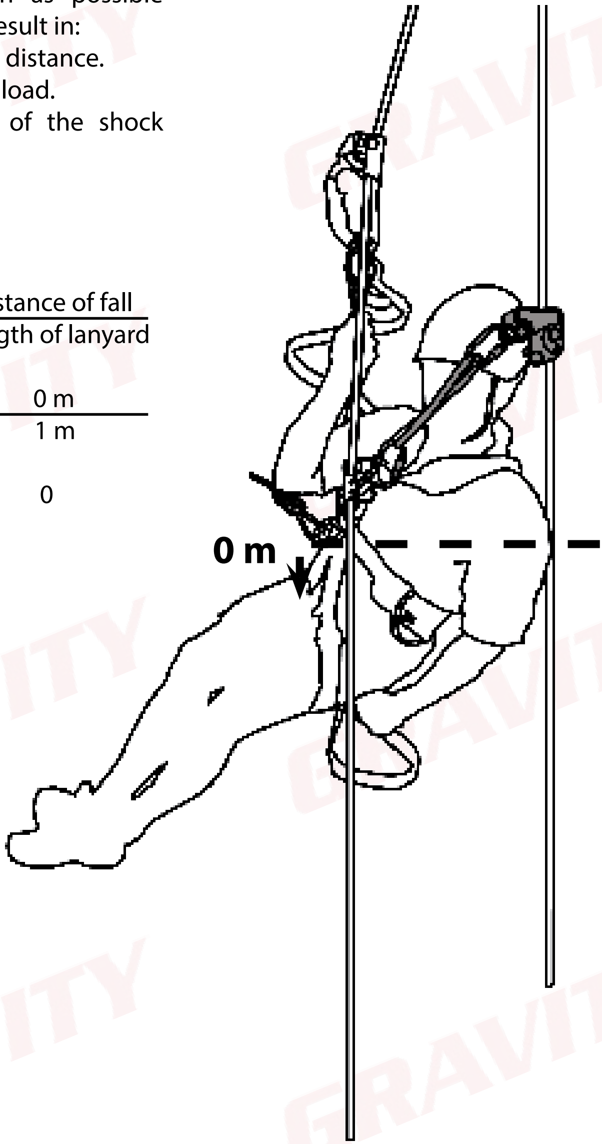
This is the ideal position to be in. Hooking up as high as possible above the head will result in:

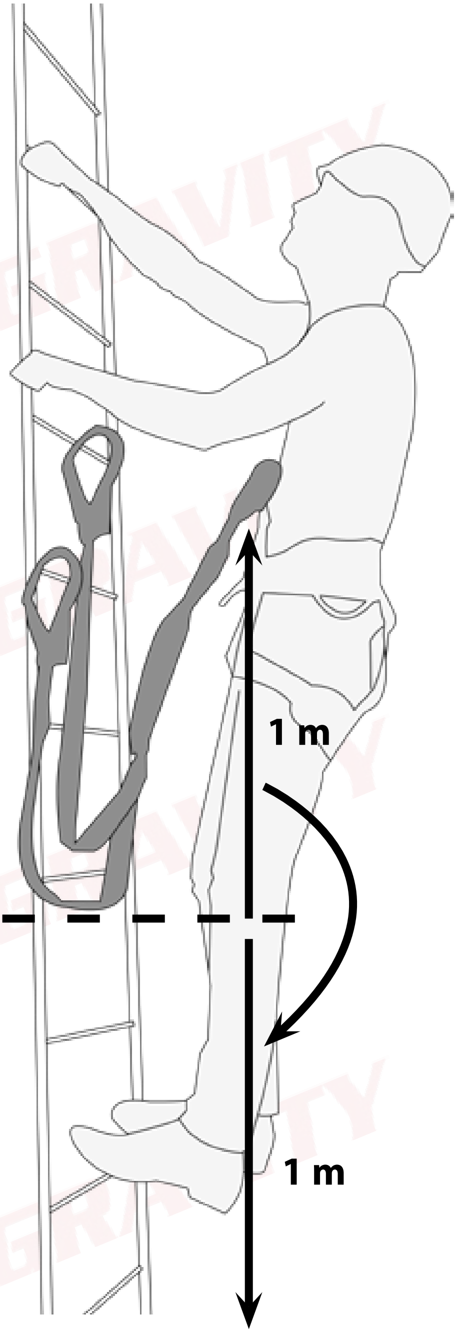
- Almost no falling distance.
- Negligible shock load.
- No deployment of the shock absorber.
- No injury.

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{0\ \text{m}}{1\ \text{m}}$$

$$\text{Fall Factor} = 0$$





Fall Factor 1:

This position, hooking up at the same level as the chest is not as safe as a Fall Factor 0, but is still suitable.

Falling from this position will result in:

- Medium falling distance.
- Slight shock load.
- Partial deployment of the shock absorber.
- Possible slight injury (bruises, whip-lash etc.).

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{2\text{ m}}{2\text{ m}}$$

$$\text{Fall Factor} = 1$$

Rope Access

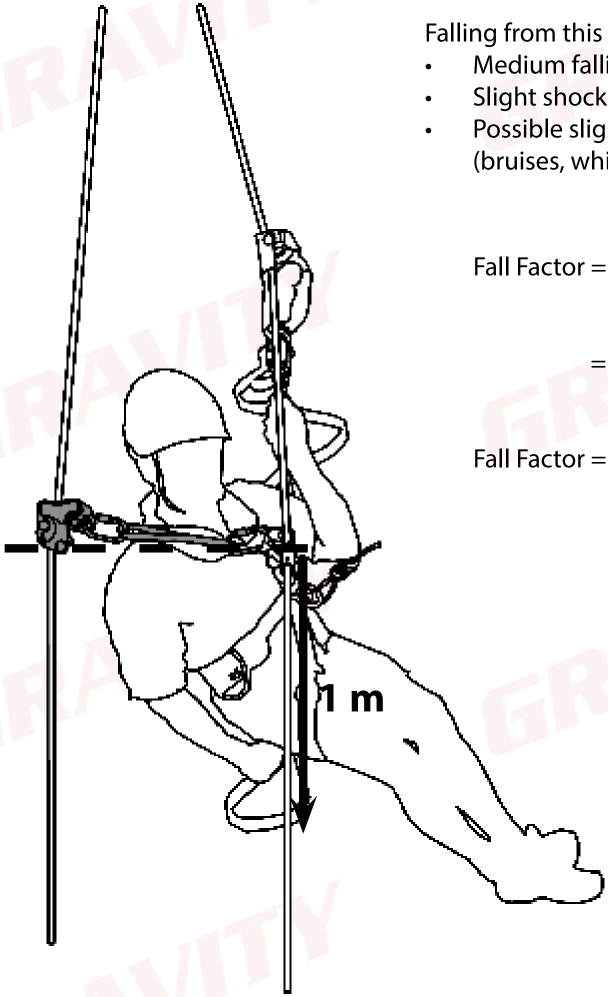
This position, hooking up at the same level as the chest, is not as safe as a Fall Factor 0, but is still acceptable.

Falling from this position will result in:
- Medium falling distance.
- Slight shock load.
- Possible slight injury (bruises, whiplash etc.).

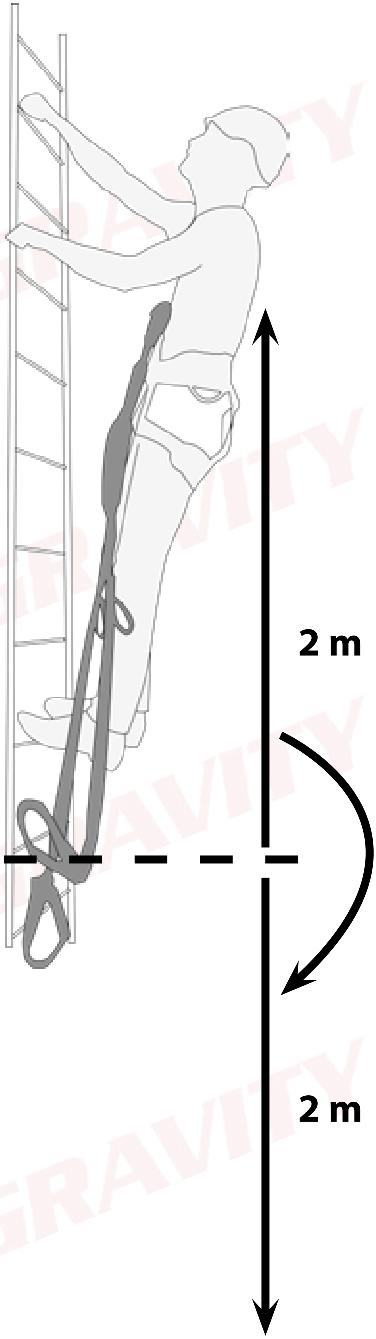
$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{1\text{ m}}{1\text{ m}}$$

$$\text{Fall Factor} = 1$$



Fall Factor 2:



This is the most dangerous position to be in. No worker should ever be allowed to work from this position; if there is no higher connection point, a different fall protection system must be used.

Falling from this position will result in:

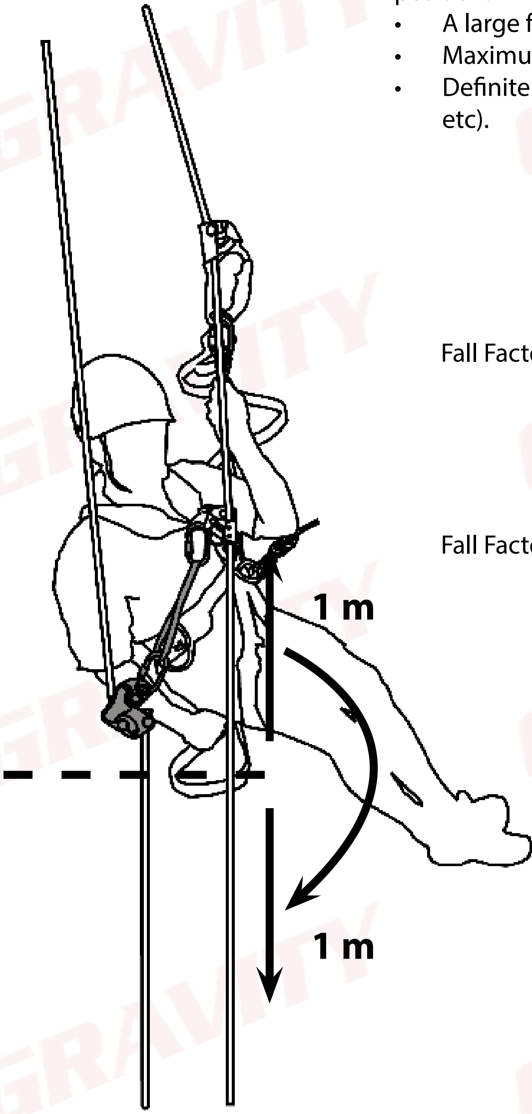
- A large falling distance.
- Maximum shock load.
- Full deployment of the shock absorber
- Definite injury (broken bones, back injury etc.).

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

$$= \frac{4\text{ m}}{2\text{ m}}$$

$$\text{Fall Factor} = 2$$

Rope Access



This is the most dangerous position to be in. No worker must ever be allowed to work from this position:

- A large falling distance.
- Maximum shock load.
- Definite injury (broken bones, back injury etc).

$$\text{Fall Factor} = \frac{\text{Distance of fall}}{\text{Length of lanyard}}$$

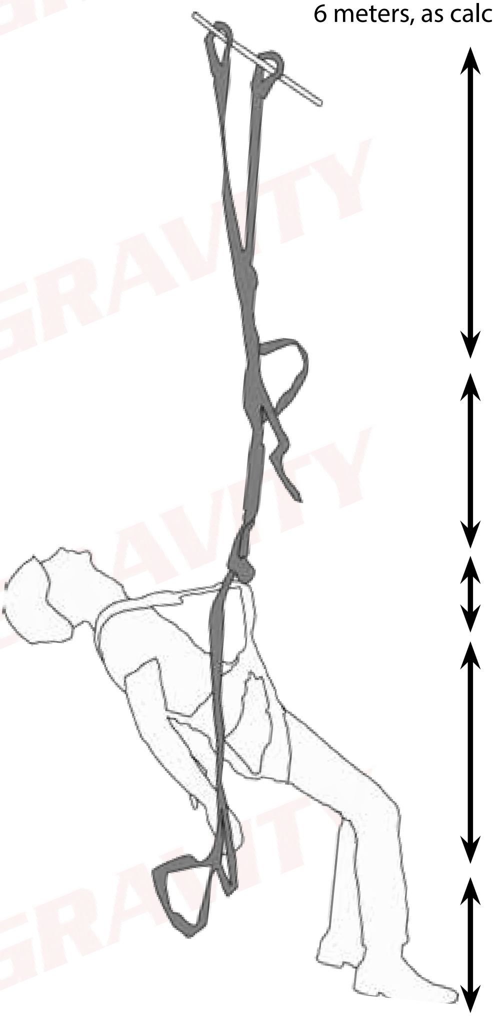
$$= \frac{2\text{ m}}{1\text{ m}}$$

$$\text{Fall Factor} = 2$$

2. Minimum Free Space

Minimum Free Space (MFS) is the open space a technician would need below him/her to safely arrest his/her fall without hitting the ground or a structure. Reducing the fall distance or Fall Factor would reduce the required MFS.

For example if a person working in a Fall Factor 2 with a 2 m lanyard falls from a structure, the minimum free space required to safely arrest a person's fall would be 6 meters, as calculated below:



2 m for the lanyard.

1.5 m for the lanyard since, at Fall Factor 2, the lanyard will deploy completely.

0.5 m for the harness stretch.

1.5 m for the persons body.

0.5 m free space so that the person does not hit the ground.

THUS, 2 m+1.5 m+0.5 m+1.5 m+0.5 m = 6 m

E. Anchor Points and Lifelines

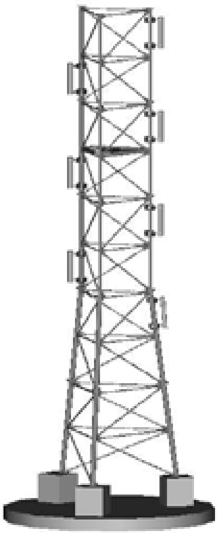
Anchor Points

Anchor points are mainly divided into two categories:
- Self-identified anchor points, and
- Planned anchor points.

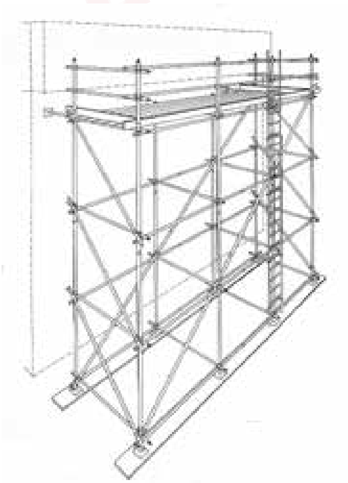
1. Self-Identified Anchor Points

Self-identified anchor points form part of the structure, meaning that should a specific component be removed, the entire structure will become weaker.
Note: Self-identified anchor points are not marked or certified as such.

Safe Working Load : 150 kg
Breaking Strength : 1500 kg



Lattice Mast



Scaffolding

A fall arrest system can be connected to any part of the structures above.
Beware of pinch-points and places where connectors can cross-load.

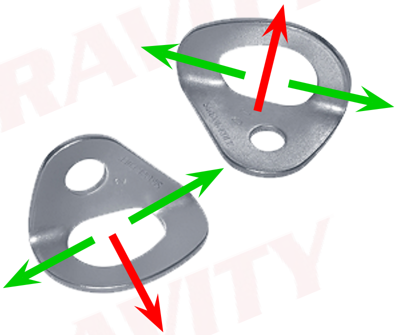
2. Planned Anchor Points (EN 795)

Planned anchor points do not form part of the structure and are placed there by a competent person/company for the specific purpose of anchoring a technician while working at height.

All planned anchor points must be **tested annually** to ensure that they are still "safe for use and fit for purpose".

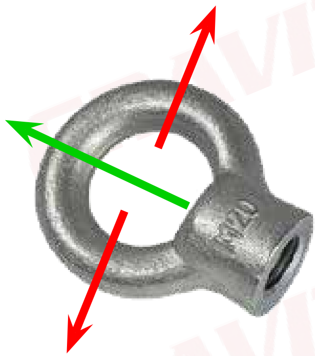
Planned anchor points must be certified and marked as such with the safe working load and the date of last and next inspection.

Safe Working Load : 220 kg
Breaking Strength : 2200 kg



Hangers

Note: Hangers can only handle a force parallel to the surface to which they have been attached to; we call this a "shear" force. Hangers can only be used in shear.



Eye-Nut

Note: An eye-nut can only handle a force perpendicular to the surface it has been attached to; we call this a "tension" force. An eye-nut can only be used in tension.

EVEN THOUGH A PLANNED ANCHOR POINT HAS A SAFE WORKING LOAD OF 220 KG, THE MINIMUM SAFE WORK LOAD OF ANY ANCHOR POINT IN A BASIC FALL ARREST SYSTEM REMAINS 150 KG.

Lifelines

Lifelines are mainly divided into two categories:

- Permanent lifelines, and
- Temporary lifelines.

Permanent Lifeline

3. Permanent Lifelines

Permanent lifelines are used for easy and quick movement up and down a structure. These lifelines are used with a guided type fall arrester/glider.

Permanent lifelines do not form part of the structure and are installed by a competent person/company for the specific purpose of securing a technician while work at height.

All permanent lifelines must be tested annually to ensure that they are still "safe for use and fit for purpose".

Temporary Lifeline

4. Temporary Vertical Lifelines

Vertical lifelines are used for easy and quick movement up and down a structure. These lifelines are used with a rope grab device which results in a reduced falling distance, only being limited to the amount of rope stretch.

Temporary lifelines are always made from rope and may be installed by a person with a US 229995 qualification.

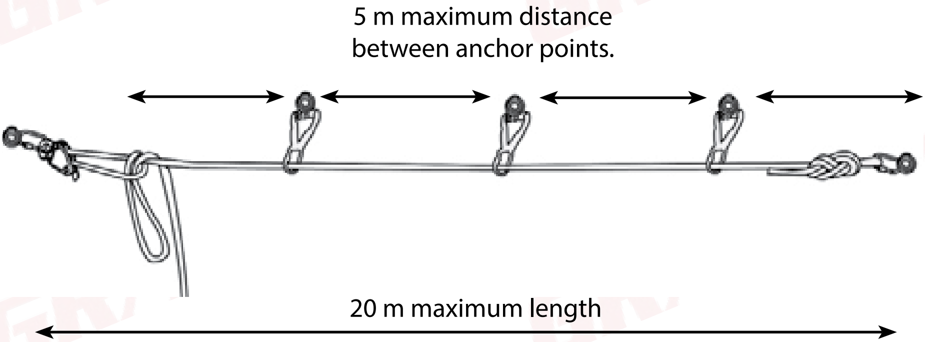
- Maximum length: 30 m
- Maximum persons allowed on the lifeline: 1 Person.

Both temporary lifelines and permanent lifelines are either vertical or horizontal.

Temporary Horizontal Lifelines

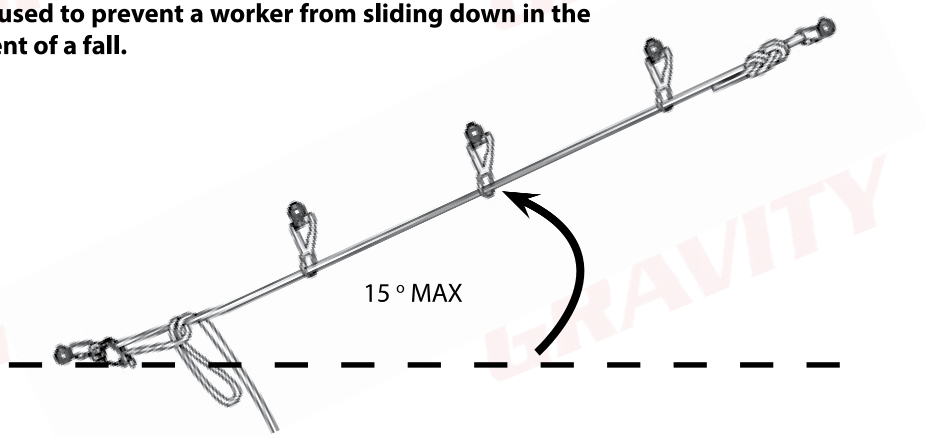
Horizontal lifelines are used for safe and quick movement from side to side. These lifelines are used with a shock absorber which can slide from side to side as the technicians moves.

A horizontal lifeline **may not be angled at more than 15 degrees** from the horizontal as this will cause a person who has fallen to slide down, and can lead to serious injury.



- Maximum length: 20 m
- Intermediate anchor point: Every 5 m
- Maximum persons allowed on the lifeline: 2 Persons
- Maximum persons per section: 1 Person

Lifelines that are angled at more than 15 degrees are considered to be vertical lifelines as a rope grab must be used to prevent a worker from sliding down in the event of a fall.

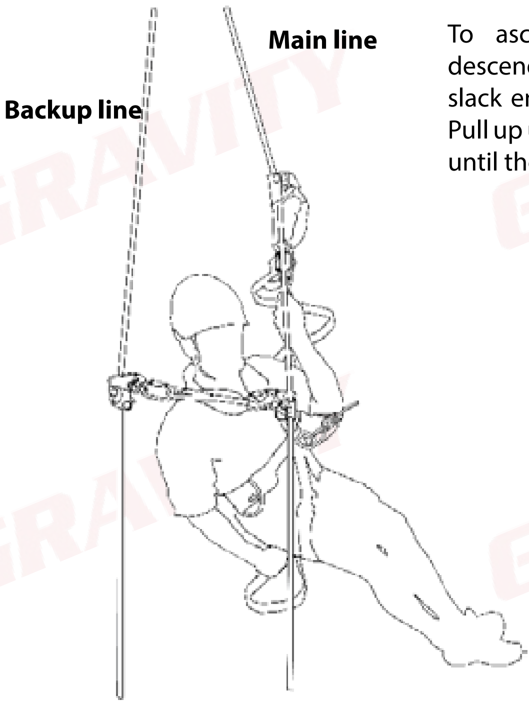


F. Rope Access Procedures

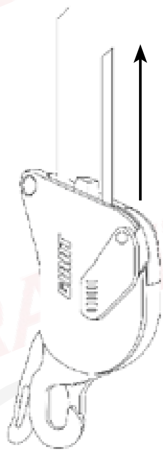
All rope access work MUST be supervised to ensure all work is conducted safely. The level of supervision required will be given per procedure. A level 3 Roep Access supervisor must conduct an assessment of the worksite in order to determine whether a site is simple enough for a Level 2 Rope Access Practitioner to supervise or whether it is of such a complexity that a Level 3 Rope Access Supervisor must be on site to provide direct supervision.

Ascending - Moving Up on a Rope

1. Install backup device on one of the ropes; this is now your backup line.
2. Install chest and handled ascender on the other rope; this is now your main line.
3. Remove stretch (if there is any) from main line.
4. Put your foot on the second step of the etrier.
5. Stand up straight whilst ensuring that the rope is passing through the chest ascender.
6. Sit down.
7. Move handled ascender upwards on the rope.
8. Ensure your backup remains in a safe Fall Factor.



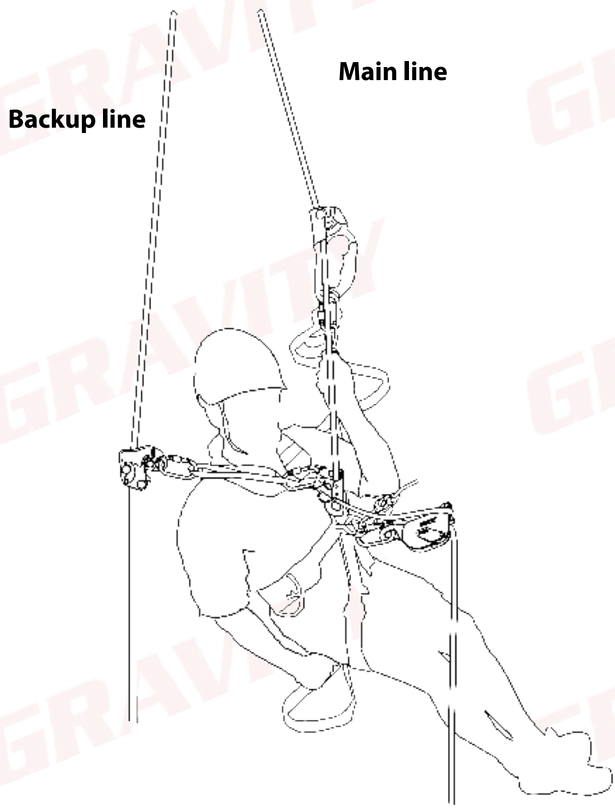
To ascend whilst connected on a descender, stand up and pull up on the slack end of the rope as shown below. Pull up until the slack is removed. Repeat until the desired height is reached.



The Change Over (ascent to descent)

When the desired height is reached, the worker must transfer their weight onto the descending device. This is in the case of an emergency so that the worker can reach the ground quicker.

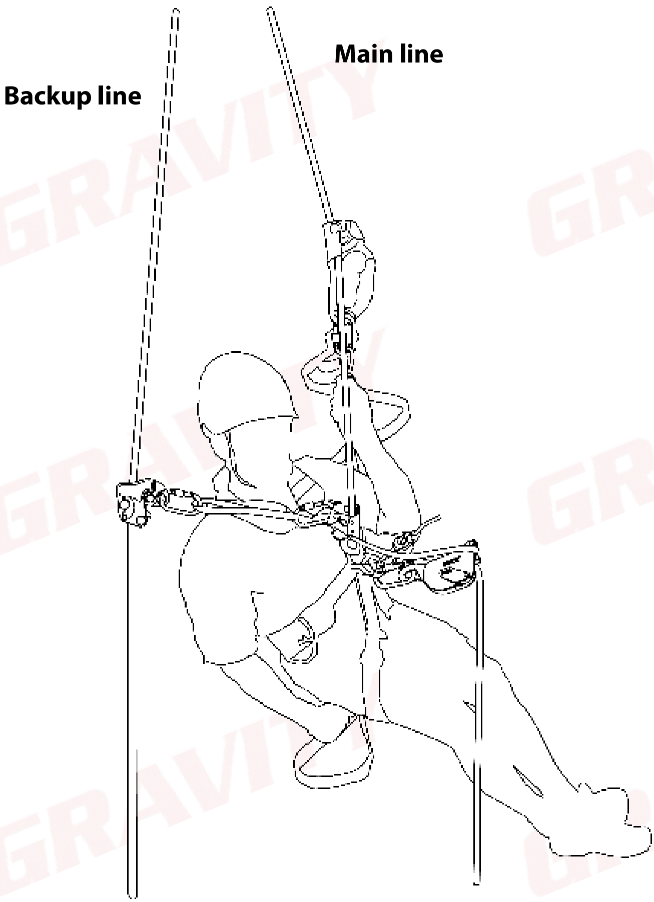
1. When the desired height has been reached, connect descender on the main line underneath the chest ascender.
2. Lock the descender.
3. Move the backup device down into a Fall Factor 1.
4. Stand up in the etrier and remove the chest ascender from the rope.
5. Sit down on the descender.
6. Remove handled ascender.
7. Descend down in a controlled fashion.



The Change Over (descent to ascent)

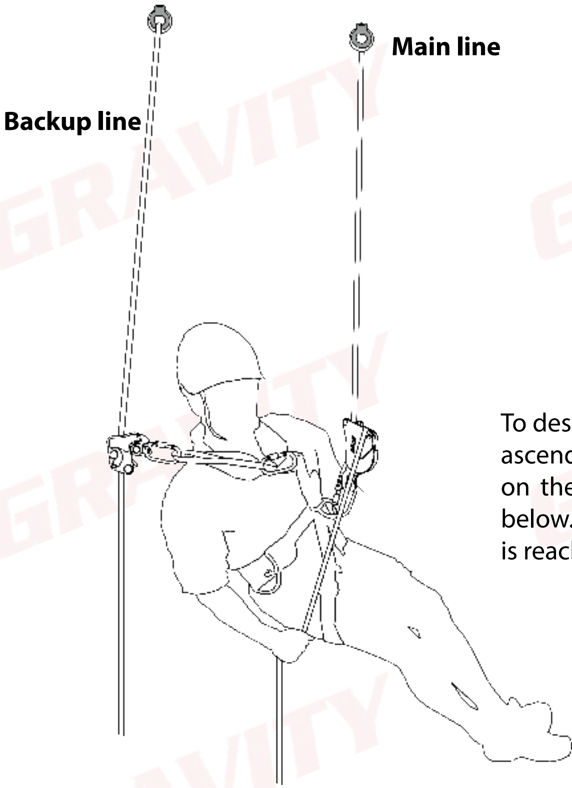
When the desired height is reached, the worker must transfer their weight onto the ascending devices. This is in the case of a worker needing to reach a higher area.

1. When the desired height has been reached, connect handled ascender above the descender on the main line.
2. Stand up and install the chest ascender above the descender.
3. Ensure that the backup device remains it at least a Fall Factor 1.
4. Remove the descender.
5. Ascend as previously described.

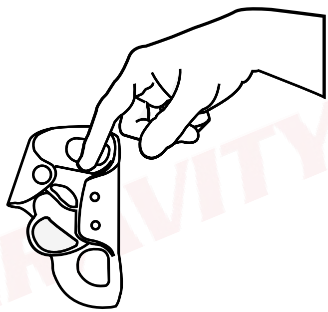


Descending - Moving down on a Rope

1. Install backup device on one of the ropes; this is now your backup line.
2. Install descender on the other rope; this is now your main line.
3. Remove stretch (if there is any) from main line.
4. Put your weight onto the main line
5. Sit down.
6. Grab the end of the rope exiting the descender by your side D-link.
7. Ensure your backup remains in a safe Fall Factor.
8. Press the handle of the descender slowly and decend in a controlled fashion



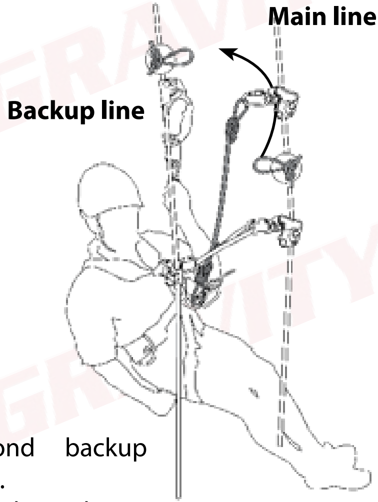
To descend whilst connected on a chest ascender, stand up and press down on the cam of the ascender as shown below. Sit down until the desired height is reached.

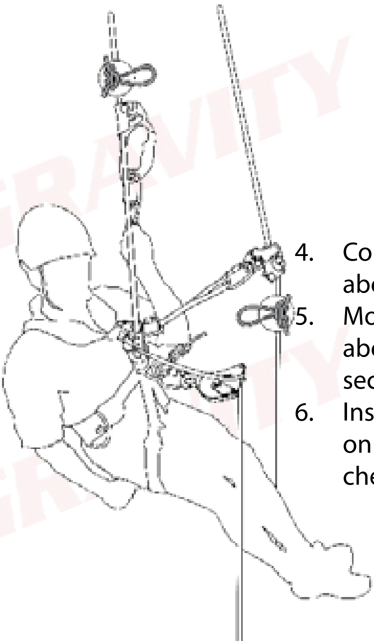


Ascending: Passing Obstruction in a Rope

There may be some obstacle in a rope for any number of reasons, e.g. knots, casualties etc. When passing such an obstruction, it is crucial that 2 points of contact are always maintained.

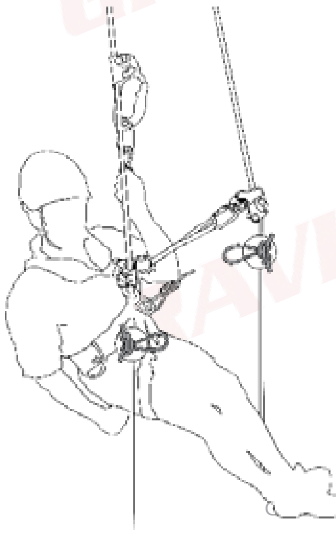
1. Connect backup device to the rope with the LOWEST knot.
2. The other rope becomes the main line.
3. Climb up until the first knot stops the backup device in a Fall Factor 1.





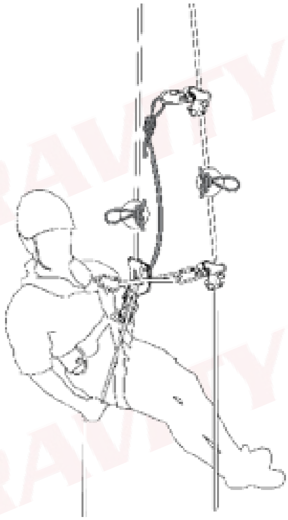
4. Connect second backup above the knot.
5. Move first backup device above knot and remove the second.
6. Install descending device on main line (underneath chest ascender).

7. Move handled ascender above knot and as high as possible.
8. In one fluid motion, stand up, open chest ascender and install above the knot.
9. Remove descender.

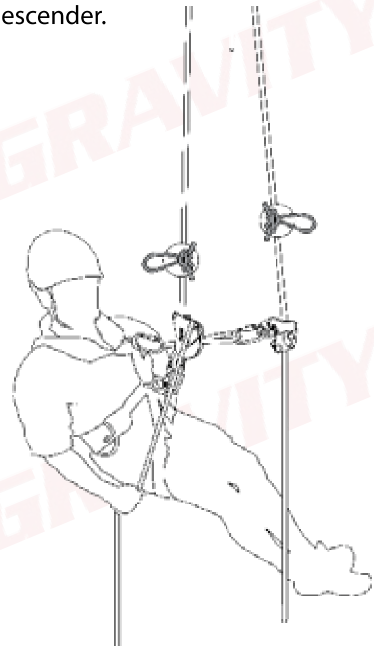


Descending: Passing Obstruction in a Rope

1. Perform a change over.
2. Descend down until the knot prevents the descender from moving down any further.
3. Install handled ascender above the descending device as high as possible on the main line.
4. Stand up in the etrier and install the chest ascender between the descender and handled descender.



5. Remove descender.
6. Install descender BELOW the knot.
7. Lock descender.



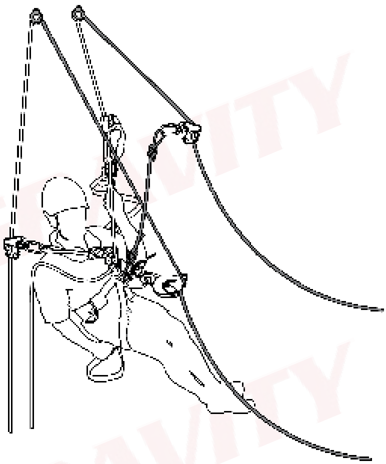
8. Remove chest ascender.
9. Remove handled ascender.
10. In the backup line, install second back-up device above knot.
11. Remove first backup device and install it underneath the knot.
13. Remove second back-up.
14. Descend down in a controlled fashion.

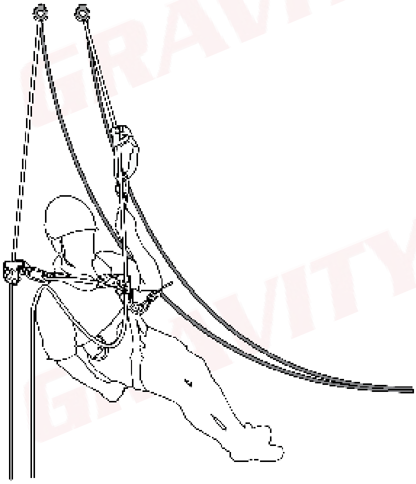
Loop

Loops can be used for horizontal movement, as well as moving underneath obstacles that may obstruct a clear path of movement, e.g pipelines or catwalks.

An extra backup device is required

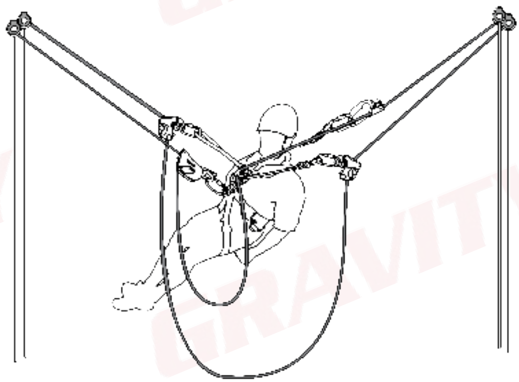
1. Ascend on the straight drop (white ropes) until the top is reached.
2. Perform a change over ONTO THE LOOP (grey ropes) .





3. The operator now has 4 points of connection.

4. Remove the ascending equipment from the straight drop.
5. The ascending gear is moved to the opposite side of the loop.
6. The loop is traversed in the same manner as the rope transfer
7. Change over onto the opposite straight drop.

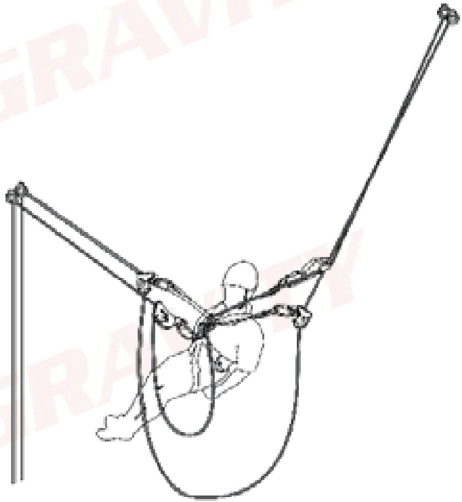


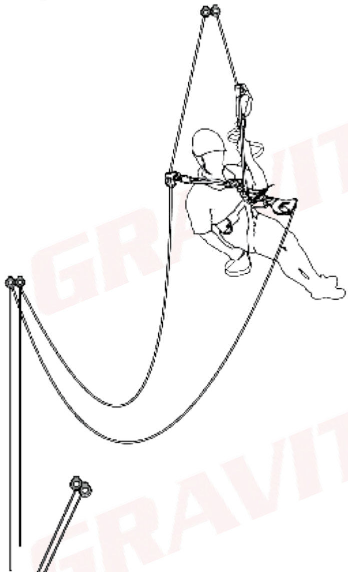
Re-belay/re-anchor

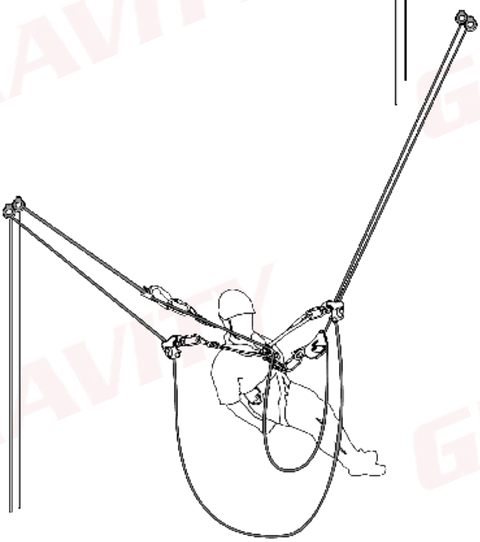
A Re-belay is the same as a loop, but has one set of straight drops rather than two.

The Re-belay is traversed in the same way as the loop with one exception:

- When the far side of the re-belay is reached, perform a change over and return through the re-belay as normal.





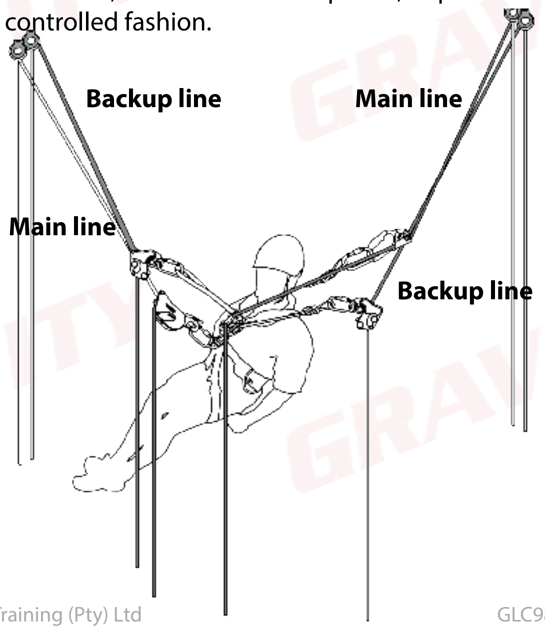


Rope Transfer

It might be required of a rope access technician to move not only up and down but sideways as well, or it might simply be more convenient to ascend on one set of ropes and descend on another. One of the methods that can be used is a rope transfer.

Before attempting a rope transfer, ensure that the two sets of ropes are connected with a secure knot and that you have a second backup device with you.

1. Ascend to at least two thirds of the height of the far away set of ropes.
2. Perform a change over.
3. Pull ropes towards you.
4. Install second backup device on one of the ropes that you have pulled towards you, ensuring that it faces in the correct direction.
5. Install the chest ascender and handled ascender on the other rope that you have pulled towards you (at this point the extra backup device, chest ascender and handled ascender should look the same as if you were going to climb straight up the second set of ropes).
6. Ascend as far as you can until it becomes difficult.
7. Descend down until the angle between the ropes is relatively small.
8. Repeat steps 6 and 7 until all of your weight has been transferred onto the second set of ropes.
9. Remove all equipment from the original set of ropes.
10. Now you can either ascend, should the task require it, or perform a change over and descend in a controlled fashion.

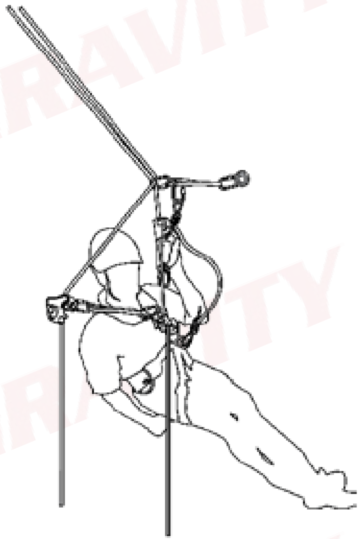


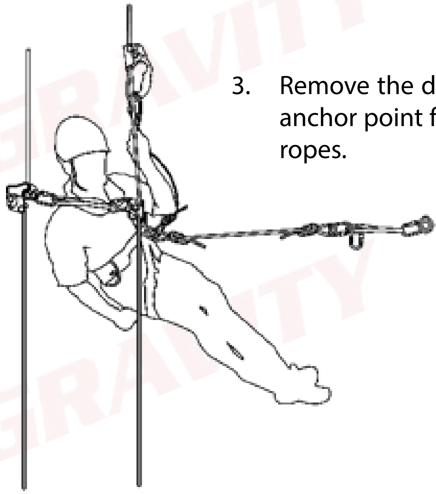
Passing a Single Deviation Up

A deviation is used when the ropes cannot hang straight down due to some hazard or lack of safe working space. The ropes are rigged in such a way that the operator need not stand directly underneath the anchor points.

A KNOT MUST BE TIED IN THE BOTTOM OF THE ROPES

1. Ascend up to the anchor point.
2. Connect an extra cowtail to the anchor point.



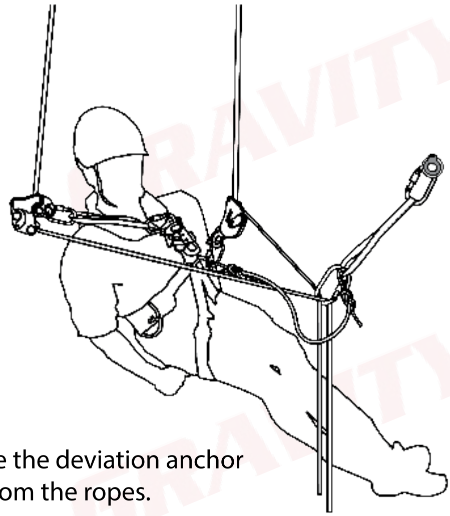


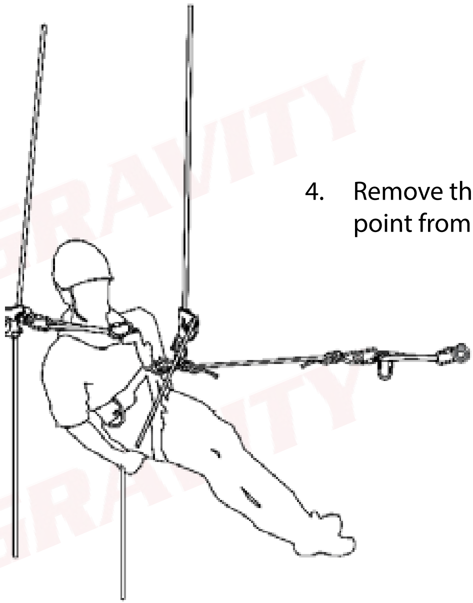
3. Remove the deviation anchor point from the ropes.

4. Re-install the deviation anchor point below the ascending equipment and backup device.
5. Ascend further.

Passing a Single Deviation Down

1. Descend down until eye level with the anchor point and lock the descender.
2. Pull yourself in using the ropes with the knot in.
3. Connect an extra cowtail to the anchor point.





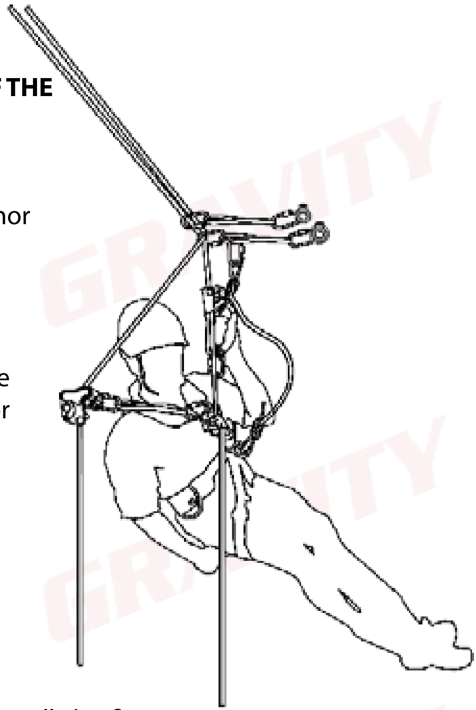
4. Remove the deviation anchor point from the ropes.

5. Re-install the deviation anchor point above the descending equipment and backup device.
6. Descend further.

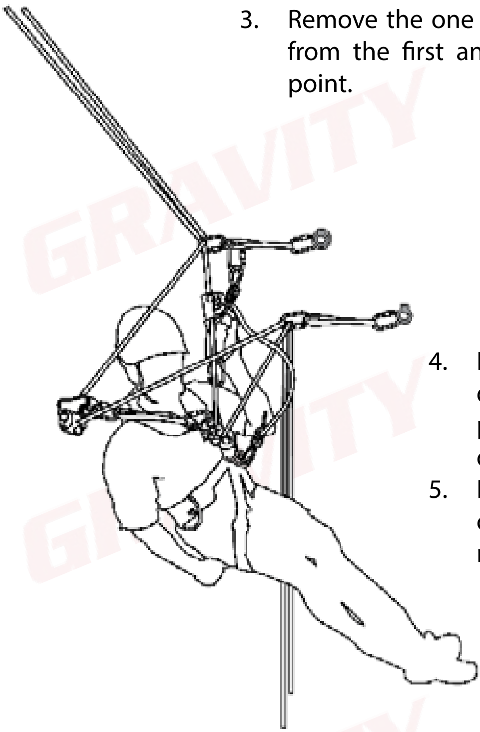
Passing a Double Deviation Up

A KNOT MUST BE TIED IN THE BOTTOM OF THE ROPES

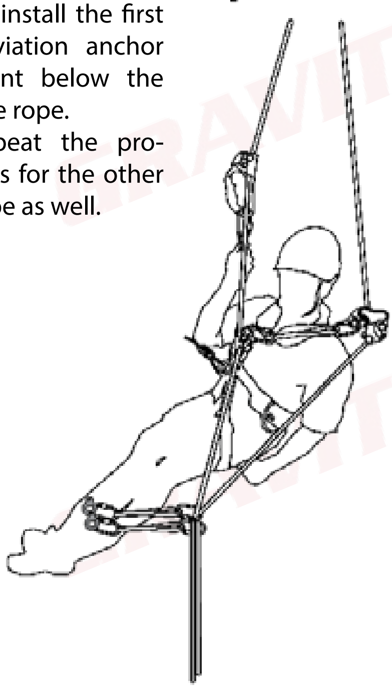
1. Ascend up to the anchor point.
2. Connect an extra cowtail to the first anchor point.



3. Remove the one rope from the first anchor point.

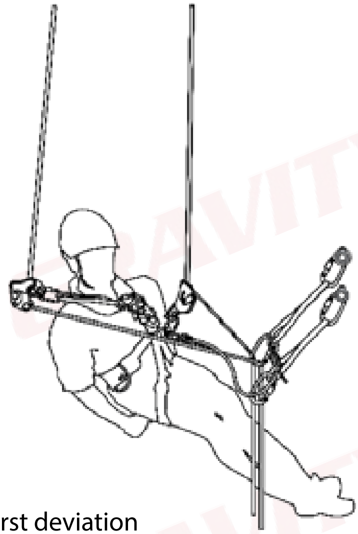


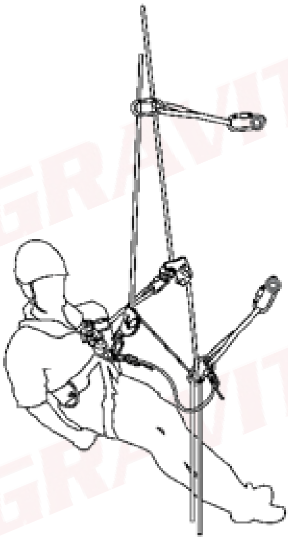
4. Re-install the first deviation anchor point below the one rope.
5. Repeat the process for the other rope as well.



Passing a Double Deviation Down

1. Descend down until eye level with the first anchor point and lock the descender.
2. Pull yourself in using the ropes with the knot in.
3. Connect an extra cowtail to the anchor point.





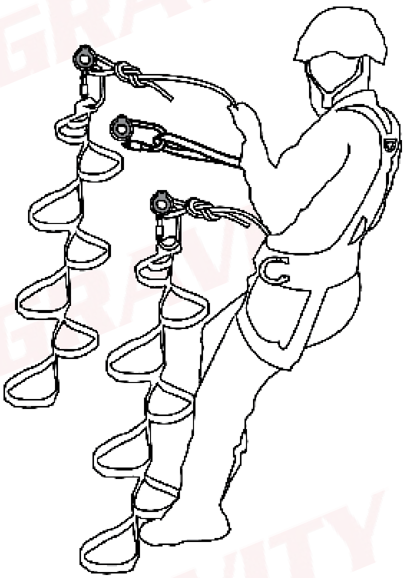
4. Remove the first deviation anchor point from the ropes.

5. Re-install the deviation anchor point above the descending equipment and backup device.
6. Descend further.
7. Repeat the process for the other rope.

Aid Climbing

Aid climbing is used to move vertically or horizontally when no ropes are available.

Sometimes there might not be ropes available on certain parts of the site. If, however, there are anchor points, one can do aid climbing to move either up or down, or side to side while at height.



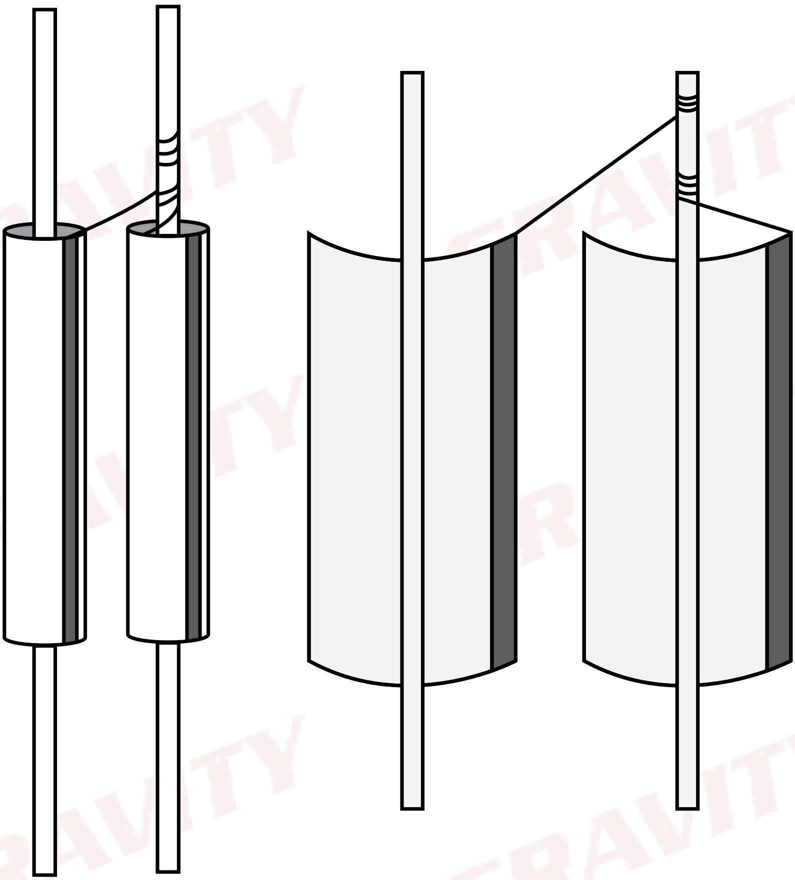
Vertical Aid Climbing for IWH only

The most important thing to remember is that you always need two points when doing rope access, and aid climbing is no exception. However, when aid climbing, each anchor point counts as 1 point. So even if you have 7 connections to one anchor point, it remains one point of connection. Thus:

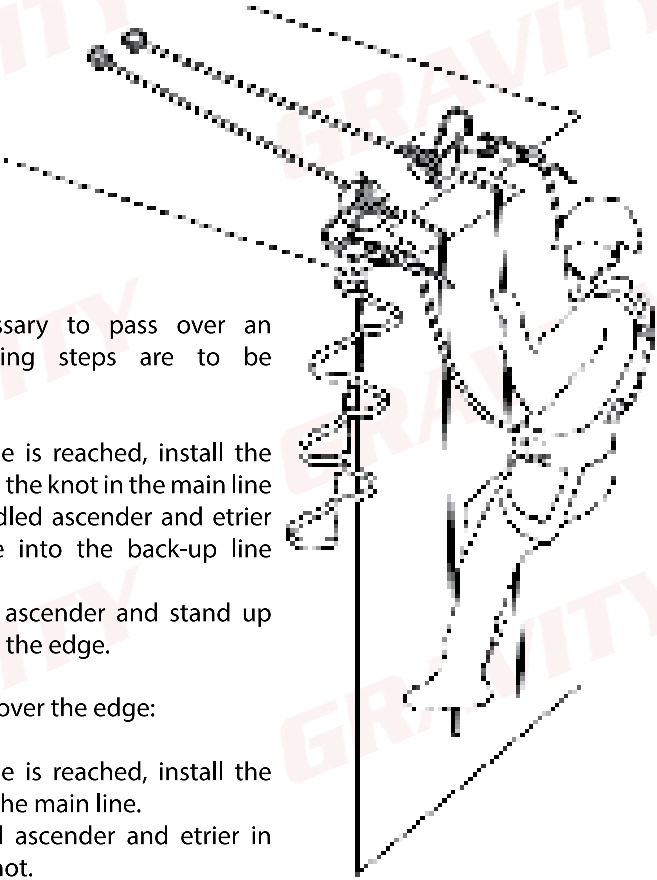
1. Connect the 2 cowtails with etriers to two separate anchor points.
2. Connect the 3rd cowtail to the lowest anchor point.
3. Stand up in the highest etrier and connect the chest ascender to that cowtail.
4. Move the other cowtail with an etrier to an anchor point above the cowtail you are currently hanging on.
5. Move the cowtail WITHOUT an etrier one anchor point up.
6. Repeat steps 3 to 5.

Passing Mid-rope Protection

1. When mid-rope protection is reached (ascending or descending), ensure that both of the protection sleeves are secured on the back-up line.
2. Open the protection and climb past the protection as required.
3. When the securing line is reached jump your back-up device as per any chosen means (similar to pass the knots).
4. Ensure that the protection is replaced exactly as it was before continuing further.



Passing Over an Edge



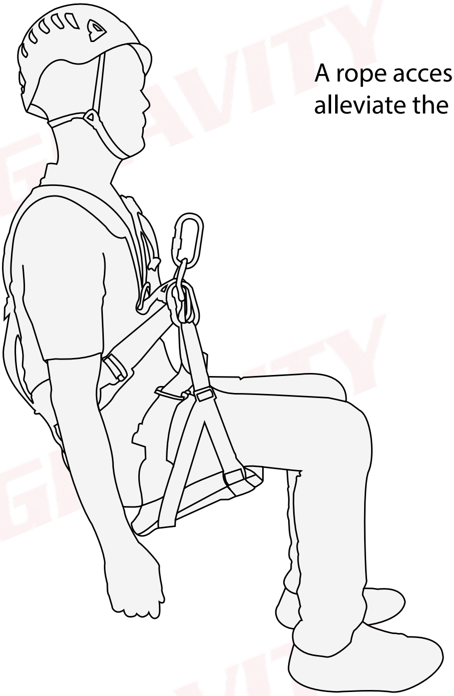
When it is necessary to pass over an edge, the following steps are to be followed:

1. When the edge is reached, install the extra cowtail in the knot in the main line
2. Move the handled ascender and etrier over the edge into the back-up line knot.
3. Remove chest ascender and stand up and move over the edge.

When Descending over the edge:

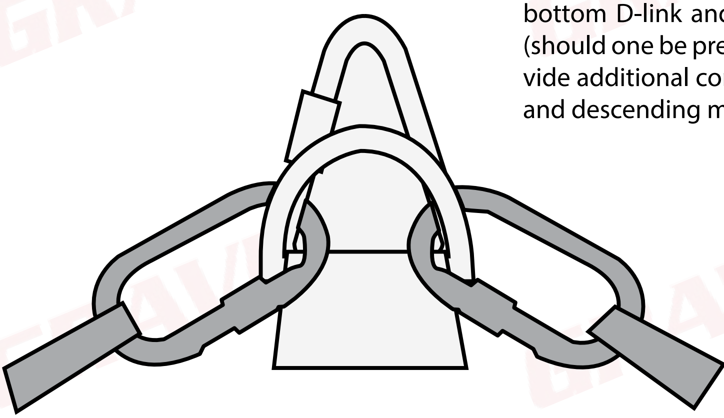
4. When the edge is reached, install the descender on the main line.
5. Install handled ascender and etrier in back-up line knot.
6. Stand up and move over the edge, and put weight onto the descender.
7. Install a backup device on the backup line.
8. Remove handled ascender and etrier.

Rope Access Seat



A rope access seat may be used to increase comfort and alleviate the symptoms of suspension trauma.

The connections of the rope access seat must be connected through the bottom D-link and the Delta Mailon (should one be present). This will provide additional comfort in ascending and descending mode.

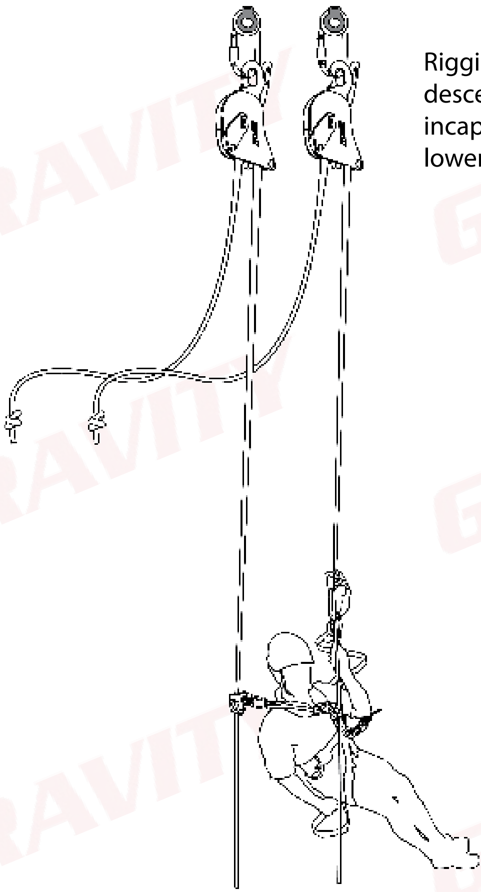


G. Emergency Procedures

ALWAYS CONTACT EMERGENCY SERVICES BEFORE STARTING RESCUES

1. Rig for Rescue

Although there are various rope access rescues that allow for a rescuer to retrieve casualties from any position imaginable, prevention is always better than the cure. Therefore, the best solution for a rescue plan is to "rig for rescue". In other words: to rig the work environment in such a way that a worker who is in need of rescue can be lowered or lifted to safety as quickly as possible, and with as little effort and as little man power as possible. For example:



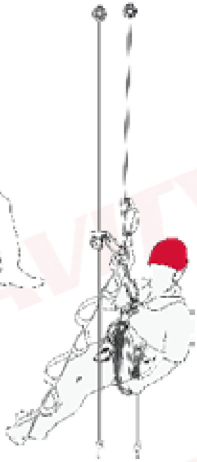
Rigging work ropes through descending devices so that a incapacitated worker can be lowered easily.

2. Rescue from a Descending Device from Separate Ropes

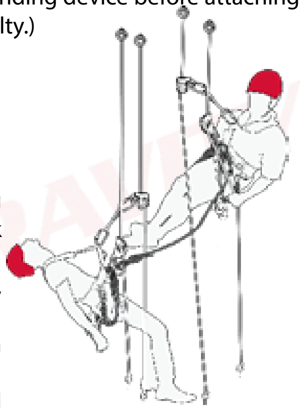
Should a worker, for any reason, be unable to reach the ground whilst on a descending device, a rescue must be performed.

You will need:

- A rescue sling
- 2 x Karabiners
- An extra locker (to be connected to a cowtail):



1. Before initiating any rescue operation, the rescuer must assess the surroundings for potential hazards and, if the situation is severe, promptly contact emergency services.
2. The rescuer should set up and rig a separate set of drop ropes as close to the casualty as possible.
3. Next, the rescuer must descend or ascend to reach the casualty's level.
4. After reaching the casualty, the rescuer must assess their condition and administer life-saving treatment if necessary.
5. The rescuer will then attach a cowtail to the bottom D-Link of the casualty. (Always ensure that you are connected to a descending device before attaching a cowtail to the casualty.)



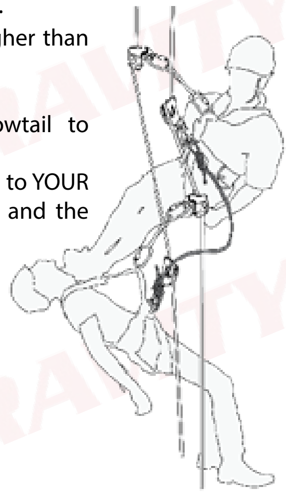
6. Subsequently, the rescuer should connect a rescue sling from the rescuers descender carabiner to the top D-Link of the casualty.
7. With the casualty connected to two points on the rescuer, the backup device of the casualty can be removed.
8. The rescuer can then lower the casualty to suspend them securely on to the rescuer.
9. The descender should be removed from the casualty and connected below the rescuer's backup device, securing it to the rescuer's harness next to the other descending device.
10. The rescuer can now remove their backup device.
11. With both the rescuer and the casualty suspended on the same set of ropes using two descenders, apply additional friction to control the descent speed, and lower the casualty to a safe location.
12. Following the rescue, the rescuer must escape the system and administer first aid procedures if necessary.

2. Rescue from a Descending Device From Same Ropes

Should a worker, for any reason, be unable to reach the ground whilst on a descending device, a rescue must be performed. You will need:

- A rescue sling
- 2 x Karabiners
- An extra locker (to be connected to a cowtail):

1. Connect your backup device on the casualty's main line.
2. Ascend until you are the same level as the casualty.
3. Connect your cowtail locker above the casualty's stop.
4. Connect your locker sling to the locker above the casualty's stop and remove your cowtail.
5. Connect the casualty's cowtail to his/her locker and remove their locker sling.
6. Ascend until you are higher than the casualty.
7. Perform a change over.
8. Connect your extra cowtail to their BOTTOM D-LINK.
9. Connect the rescue sling to YOUR DESCENDER KARABINER and the CASUALTY'S TOP D-LINK.



10. By controlling the casualty's descender, slowly descend the casualty until the descender becomes slack.
11. Remove the casualty's descender from the casualty's harness, BUT NOT FROM THE ROPE.

12. Connect the casualty's descender to your harness; ENSURE THAT THE GATES OF BOTH KARABINERS ARE FACING DOWNWARD AND TOWARDS YOU, and remove all the slack and lock it.
13. Remove your backup device.
14. Use the karabiner connecting the casualty's locker sling to their harness and connect this karabiner to the slack ends of BOTH ropes.
15. Whilst keeping tension on the slack ends of the rope running through the extra karabiner, descend down in a controlled fashion.
16. Open both karabiners and remove your BOTTOM D-LINK.

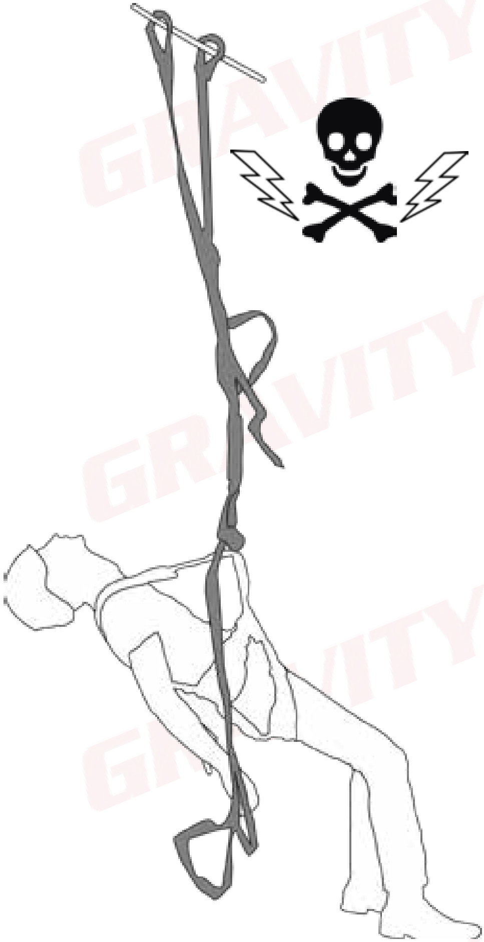
3. Suspension Trauma

The two biggest risks associated with awaiting a rescue while suspended in a harness, are injury or death due to suspension trauma.

Suspension trauma is a serious risk and can possibly be fatal if prevention techniques are not correctly applied.

Due to the force of gravity, the blood in the fallen person's body is pulled toward the ground, and pools in the legs. The average person has approximately 6-7 liters of blood in their body, and each leg can accommodate up to 6 liters of blood, thus almost all of your blood can fit into one leg.

Due to the constriction of the leg loops of the harness, the blood can not return to the fallen person's heart. This prevents blood being circulated through a person's body. As blood pooling in the legs occur, less and less blood reaches the brain, leading to loss of consciousness and can also lead to brain damage should it be left unchecked. After this the blood levels in the heart will start to lessen, causing the heart to have little to no blood to circulate through a person's system. Finally, this can lead to cardiac arrest and eventually death.



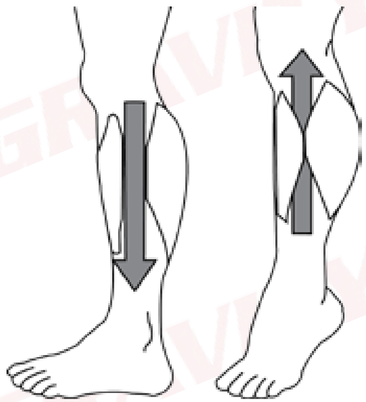
All of this can happen within 15-20 minutes; this is why it is so important to have a good knowledge of the techniques needed to prevent suspension trauma, for yourself, and the casualty that needs to be rescued.

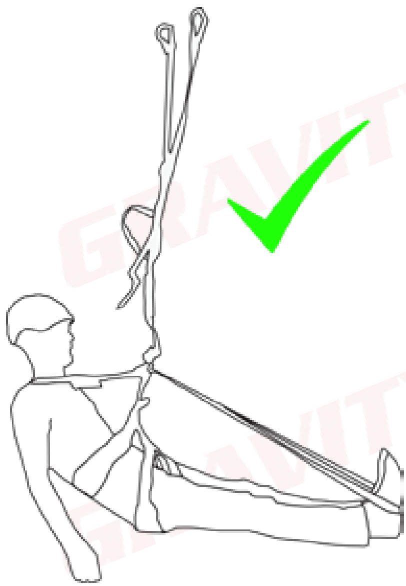
Preventing Suspension Trauma

Even though suspension trauma is very serious, a few very simple techniques will make it possible to be suspended in a harness indefinitely.

Muscle Pump: Moving your legs, feet and toes will contract the muscles in the legs and force blood back up through the leg loops, and into circulation, which provides the brain with the much needed oxygen to stay conscious.

Get Horizontal: Putting your feet up on the structure or into a suspension trauma sling will reduce the pull of gravity on the blood in your body and this will prevent the blood from pooling in the legs.





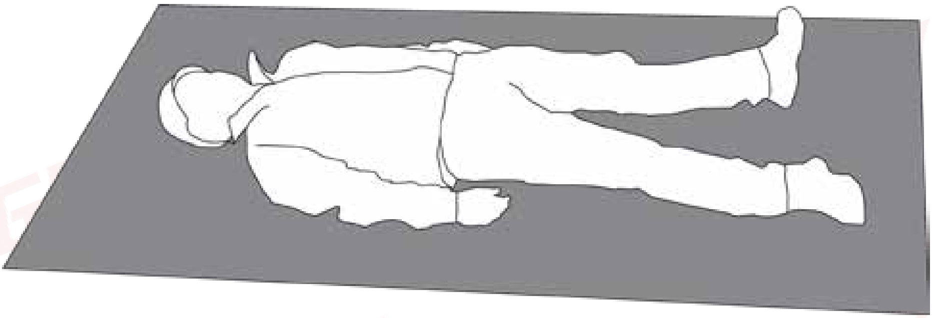
Suspension Trauma Sling: Most of the modern harnesses and shock absorbers are fitted with straps that deploy or that can be deployed as needed when suspended in a harness. This will allow you to stand up or to get horizontal, prevent the blood from pooling in the legs as well as relieve the pressure of the leg loops which will allow the blood to remain in circulation to the heart and brain. Should the casualty be unconscious, the rescuer can do this for the casualty.

Beware! After a rescue of a person who has been suspended for a while has been completed, the blood that was not in circulation will have become toxic, and if this passes through the body too quickly it can lead to a heart attack. This is referred to as **Reflow Syndrome.** Thus, take care not to move a person, or allow them to move too vigorously after a rescue has been completed.

After a casualty has been lowered down to a point of safety, it is the duty of the first aider to treat the casualty for any symptoms of suspension trauma, as well as positioning the casualty in the "Recovery Position".

4. Recovery Position

This position allows the blood to flow slowly enough as to prevent Reflow Syndrome, while at the same time keeping the casualty stable and still as to prevent any possible further injury to the casualty's neck and/or spine. This position is also known as the "Supine Position".



H. A Guide to Assessment

The following is a non-exhaustive list of **major** discrepancies (Fail):

a) Only one point of safety attachment while in suspension;
b) Unable to complete the task;
c) Excessive time period;
d) No backup to protect against a potential out-of-control swing that may cause injury or damage in the event of failure of an item of equipment;
e) Harness unsecured;
f) Anchor lanyards and device lanyards, e.g. cowtails tied or attached dangerously;
g) No helmet at height;
h) Critical harness connectors unfastened or unsecured, e.g. screw links (maillon rapides);
i) Misuse causing damage to equipment;
j) Unsuitable choice of rope protection measures;
k) Uncontrolled descent during a task or rescue;
l) Descending device threaded incorrectly and used in that manner;
m) Backup or other devices used upside down;
n) No safety attachment close to an exposed edge;
o) Excessive slack in connection to an ascending device used as a point of attachment;
p) Critical safety issues as defined by the Assessor;
q) A swing that could cause injury to personnel or damage to equipment or property.

The following is a non-exhaustive list of **minor** discrepancies (Dis):

a) Descending device not locked off or no control of the tail rope;
b) Attachment connectors not secured;
c) Critical personal fall protection equipment dropped;
d) Rope protection incorrectly placed;
e) No extra friction used when required;
f) Harness incorrectly adjusted;
g) Helmet chin strap unfastened;
h) Critical personal fall protection equipment missing from the harness set-up;
i) Tangle of ropes;
j) Poor management of backup device (major if critical);
k) Work positioning lanyards such as cowtails positioned greater than Fall Factor 1;
l) Excessive slack in connection to an ascending device used as a point of attachment (major if critical);
m) Considerable time taken to perform the task;
n) Unconventional or untrained techniques used;
o) A small out-of-control swing.

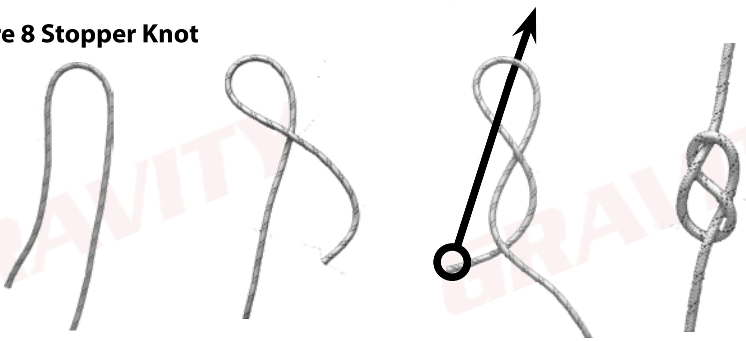
H. Knots

Use: Knots will be used extensively throughout this course, as well as in any work at height, therefore it is important to know a few important facts about knots.

- All knots weaken and decrease the length of the rope.
- All knots must be dressed properly; an incorrectly dressed knot can be up to 50% weaker than a correctly dressed knot.

Four knots will be used in this course:

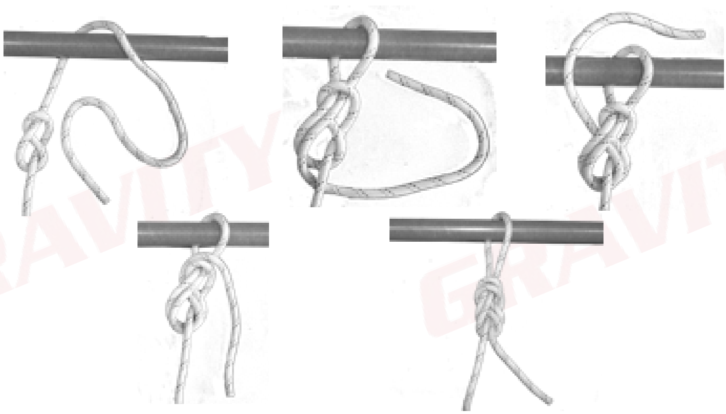
1. Figure 8 Stopper Knot



This knot is made in the end of a rope to stop people and equipment from being able of move off the end of the rope. The tail must be longer than your fist, but shorter than your forearm.

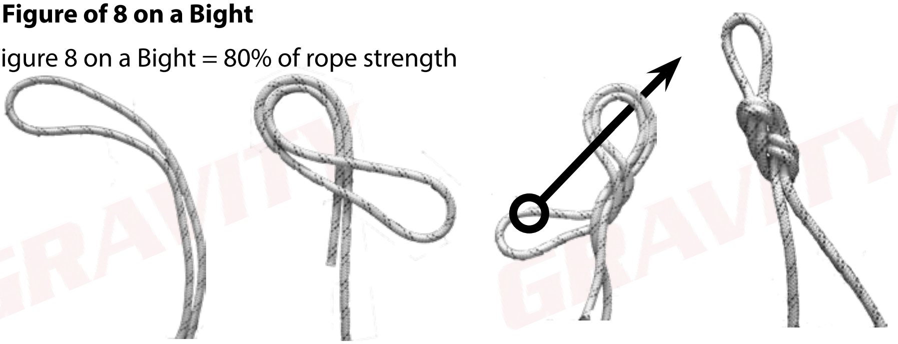
2. Figure 8 Follow Through

Simply put, this is a stopper knot that has been completed, which leads to a Figure 8 on a Bight without a connector. This is used to connect cowtails to a harness.



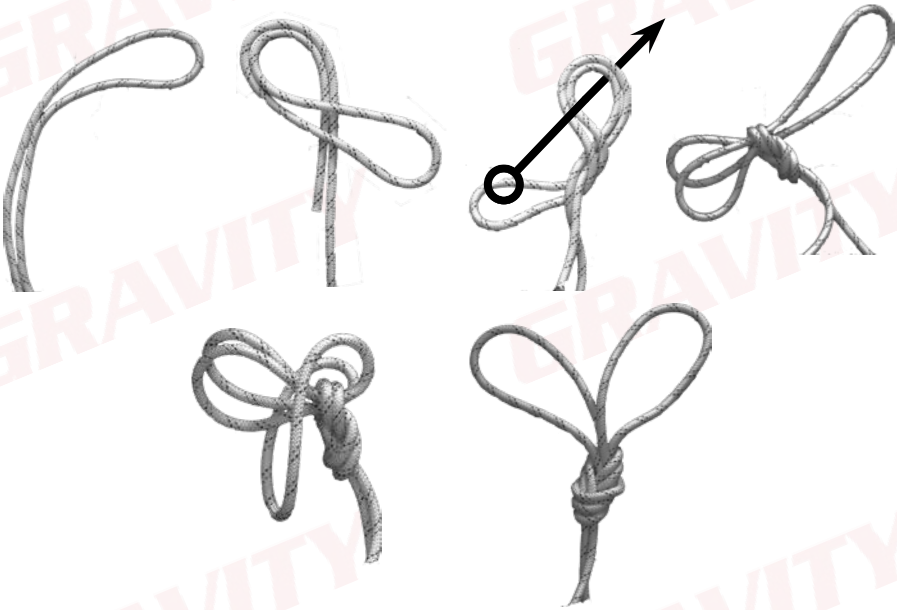
3. Figure of 8 on a Bight

Figure 8 on a Bight = 80% of rope strength



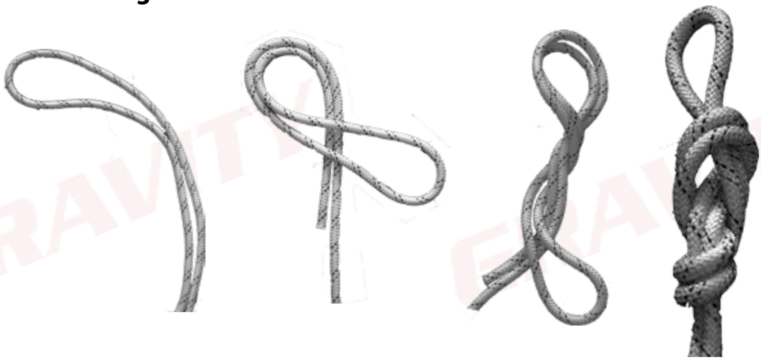
Use: To connect the rope to a single anchor point or devices, for the installation of horizontal lifelines, as well as during the rescue process. Much easier to untie than the overhand, it does not have the same tendency to jam and so injure the fiber, and is larger,stronger, and equally secure.

4. Double Loop Figure 8



Use: To connect a rope to two separate anchor points. Most commonly used to rig vertical lifelines.

5. Figure 9 on a Bight



This knot is used for similar reasons as the Figure 8 on a Bight, however the Figure 9 is better suited when lifting heavier objects as it avoids sharp bends and, therefore, the rope retains about 80% of its strength and it unties more easily after it has been loaded.

6. Klemheist

Used to create an anchor point on a vertical or diagonal pole. Must be used in a Fall Factor 1 or better.

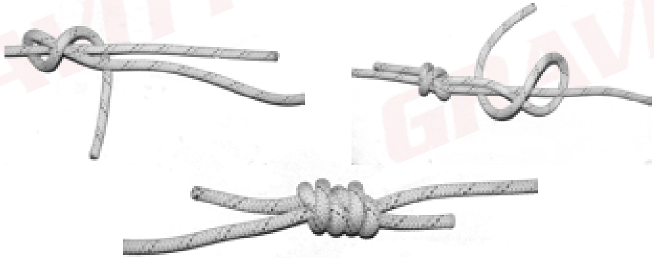
7. Barrel Knot

Use: The Barrel knot can be used self tightning loop at the end of a rope to connect the point to connectors or anchors.
A barrel knot is an adjustable knot that self-tightens when used to tie one rope, or monofilament nylon line, to anothers.

Be careful, after heavy loading, the Barrel Knot becomes very difficult to untie. The tail must also be longer than 10 cm to prevent accidental untying.

8. Double Fisherman

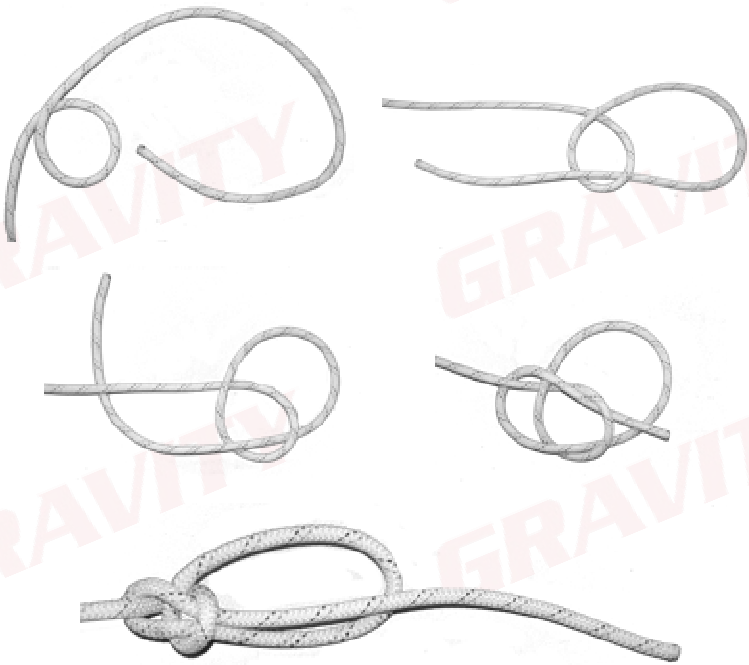
A Double Fisherman is tied exactly the same, however, the Fisherman knot is tied AROUND another rope (as shown below). It is the way to join two ends of a line to form a Prusik Loop and is also an excellent and reliable way of joining two climbing ropes

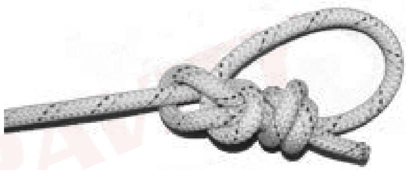


9. Bowline

Use: A knot that can be used like the Figure 9 for lifting heavier objects as it unties very easily after being loaded and is very quick to tie. It must, however, be tied off with a Fisherman knot.

- Easy to tie: The bowline is a simple and fast knot to tie, making it a good option when caving or in rescue situations.
- Resistant to slipping: Bowline knots do not slip or bind when stressed by a heavy load.



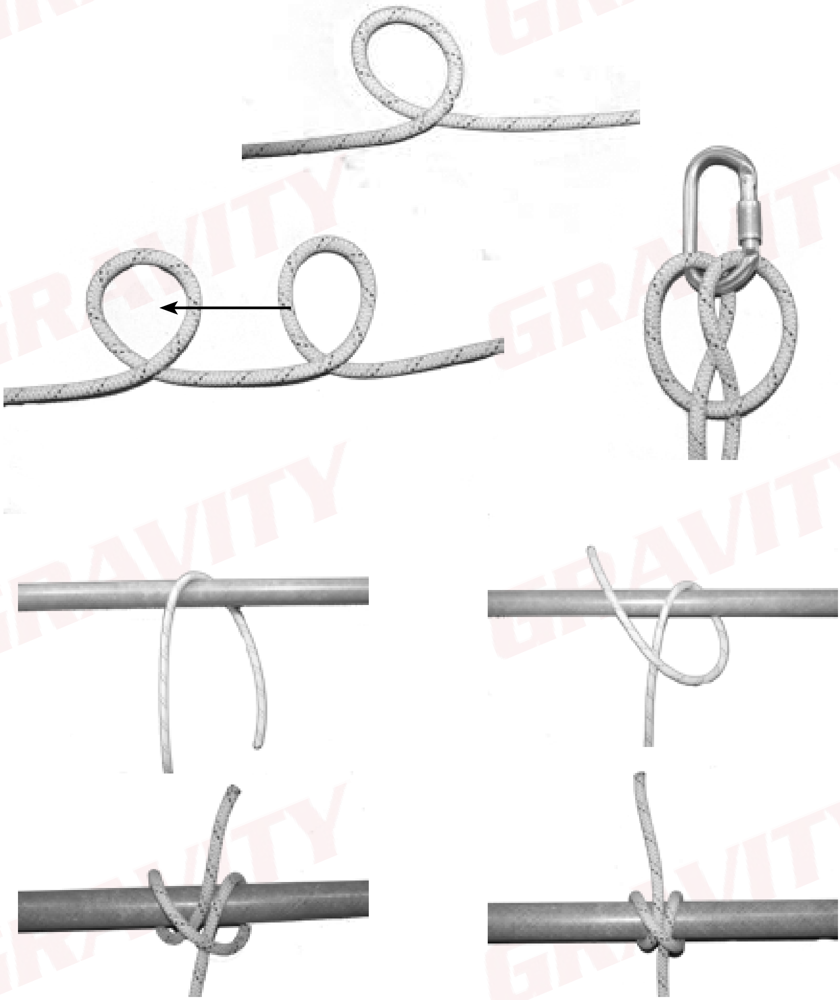


A Bowline tied off with a Barrel/Fisherman knot.

10. Clove Hitch

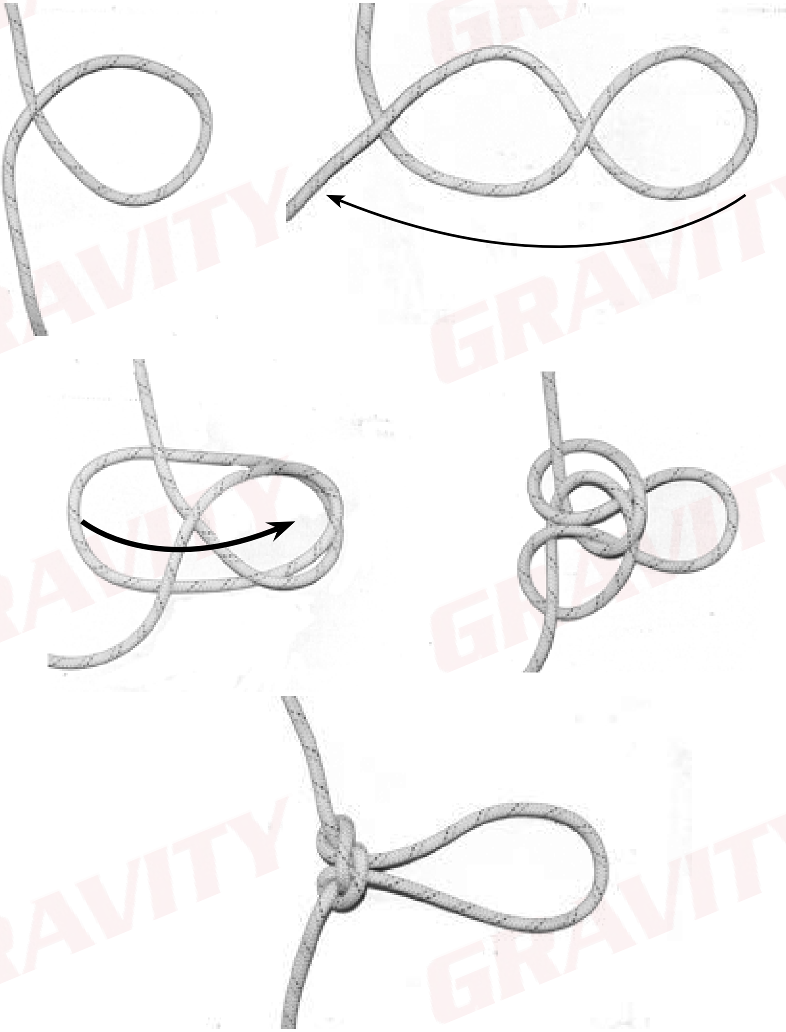
Use: The Clove Hitch is used to either create a temporary anchor point around a beam, or to create a temporary eye in the middle of the rope to lift tools etc.

The real advantage of a clove hitch is that you can quickly tie it in the middle of a rope.



11. Alpine Butterfly

Use: One of the most useful knots, the Alpine Butterfly can be used to tie into the middle of the rope, to isolate damage in a rope or to create an eye in the middle of the rope which can be pulled in three directions.



12. Tape Knot

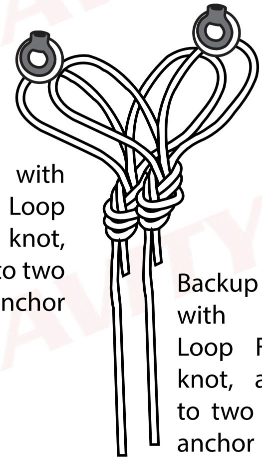
Use: A knot used to turn webbing into a sling. This knot MUST NOT be shock loaded as this may cause it to break or come loose, neither may it be used for anchoring. Remember that the tails must be at least 10 cm long.

Rigging Work Ropes

When rigging work ropes we use a double loop Figure 8 for BOTH the main line and backup line.

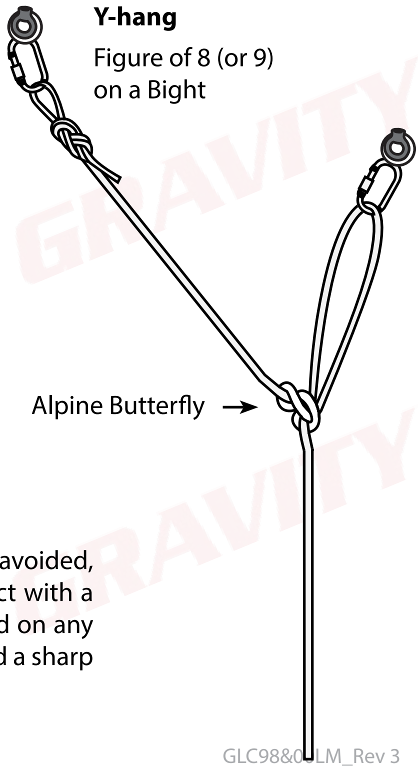
Ensure that the two loops of the same double loop Figure 8 are not connected to the same anchor point.

Main line, with Double Loop Figure 8 knot, anchored to two separate anchor points.



Backup line, with Double Loop Figure 8 knot, anchored to two separate anchor points.

Rigging two ropes with two independent anchor points with redundancy.



Ensure that all hazards such as sharp edges are avoided, by either ensuring the ropes are not in contact with a sharp edge or that edge protection is installed on any part of the rope that cannot be moved to avoid a sharp edge.

Example of an IWH Rope Access Logbook

Here are the abbreviations that may be used to describe the type of work that was done on site.

An example of a completed logbook is given on the next page.

A/D	Ascending and descending
LT	Line transfers
DEV	Deviations
L	Loop
RB	Re-belay
AID	Aid climbing
HL	Hauling and lowering systems
CW	Cable ways
B	Bolting - testing or placing bolts/anchor points in accordance with SANS 50794:1996
TR	Training - Any training that involves performing rope access manoeuvres.
ER	Emergency rescues - performing of rescues

Date: 2018.01.20-25

Work site location: Kusile Powerstation

Time arrived on site: 08:00

Time leaving on site: 17:00

Working height: 50m

Actual hours worked: 7

Employing company: ABC Maintenance

Job details: *tick applicable options

A/D X	LT	DEV	L	RB	AID X
HL	CW	B	ER	TR	

Job specifics: Boiler cleaning

Running total hours worked: 45

Supervisor name: John Doe

Registration number: 123456

Contact number: 081 123 456

Supervisor signature:

Date:

Work site location:

Time arrived on site:

Time leaving on site:

Working height:

Actual hours worked:

Employing company:

Job details: *tick applicable options

A/D	LT	DEV	L	RB	AID
HL	CW	B	ER	TR	

Job specifics:

Running total hours worked:

Supervisor name:

Registration number:

Contact number:

Supervisor signature:

Example of an IRATA Rope Access Logbook

WORK		
Date	Employing Company	Details of task being undertaken
02-13 Janurary 2018	ABC Construction	Ascending/descending painting
20-21 Janurary 2018	ABC Construction	Ascending/descending, window washing

EXPERIENCE					
Location	Hours Worked			Max Height worked	Supervisor’s Signature
Contruction Site XYZ Pretoria	x	5	0	30m	Sign
Reserve bank Pretoria	x	8	0	50m	Sign

THE WAY FORWARD

Congratulations on completing Gravity Training's Level 1 Rope Access course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

At Gravity, we care about the professional development of our clients. Should you wish to take the next step, consider enrolling for some of our other courses. See The Way Forward and the Gravity Skills and Development Map or consult your facilitator/assessor.

Some great ideas:

US 14706 - Rope Rigging

US 229994 - Fall Protection Plan Developer

US 229996 - Level 2 Rope Access (after 1000 signed off hours as Level 1)